

INVARIANT ARITHMETIC OF CLOSED GEODESICS

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ABSTRACT. We study rational valued counts of geodesic strings (reparametrization equivalence classes of closed geodesics) for complete Riemann-Finsler manifolds, based on the Fuller index of the geodesic flow. The main conceptual result is a product type formula for these counts. Combined with KAM theory, it yields the following new phenomenon. Let g be a generic Finsler metric on T^2 , sufficiently C^∞ -close to a flat metric, let β_0 be an indivisible free homotopy class, and let m be odd. If g has a primitive closed geodesic in the class $m\beta_0$, then it has another one. This is one half of a complete arithmetic law satisfied by the signed counts of primitive closed geodesics in the classes $e\beta_0$, which in the even classes takes the form of a period doubling constraint. We also obtain arithmetic constraints on counts of closed geodesics in mapping tori and flat bundles, and constraints on the existence of negative sectional curvature metrics. We also discuss results indicating a topological invariance phenomenon of these counts; such invariance would give analogous arithmetic laws for metrics on manifolds having the tautness property studied here.

1. INTRODUCTION

A *geodesic string* will be a short name for the equivalence class of a closed, constant speed g -geodesic up to reparametrization S^1 action. Suppose for the moment X is compact. For a nontrivial free homotopy class β of loops in X , we say that a metric g on X is β -*regular* if all of its class β closed geodesics are nondegenerate in the usual sense, or equivalently the closed orbits of the associated geodesic flow are dynamically nondegenerate. For a β -regular g , with finitely many class β geodesic strings we define:

$$(1.1) \quad F(g, \beta) = \sum_o \frac{(-1)^{\text{morse}(o)}}{\text{mult}(o)} \in \mathbb{Q},$$

where:

- The sum is over class β g -geodesic strings o .
- $\text{morse}(o)$ denotes the Morse index of o (meaning the Morse-Bott index of the associated critical submanifold (diffeomorphic to S^1) of the loop space).
- $\text{mult}(o)$ is the geometric multiplicity, equivalently the order of the corresponding isotropy subgroup of S^1 , where S^1 is acting on the loop space by reparametrization.

We study invariance properties and computations of the above F -count, and its generalizations using the Fuller index [23]. This also leads to novel applications, particularly to arithmetic constraints on counts of closed geodesics and their multiplicities. One of the principal tools is a certain product type formula for the F -count. While the Fuller index itself is a very general construction in dynamical systems, the product formula is a more subtle phenomenon which needs the Riemann-Finsler geometric context. A detailed outline of Fuller's theory and basic definitions are given here in the Appendix, so no prior knowledge is necessary.

Key words and phrases. Nonpositive curvature, Fuller index, geodesic counts, S^1 -equivariant homology of the loop space, KAM theory.

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For a free homotopy class β of loops in X , let $L_\beta X$ denote the corresponding component of the free loop space. The following is a direct corollary of Theorem 2.16, and exemplifies the basic topological invariance of the F -count:

Corollary 1.2. *Let X, g, β be as above and such that β is indivisible (cannot be represented by a multiply covered loop). Then the S^1 -equivariant homology $H_*^{S^1}(L_\beta X, \mathbb{Z})$ has finite total rank. Denote by $\chi^{S^1}(L_\beta X)$ the Euler characteristic of this homology. Then*

$$F(g, \beta) = \sum_o (-1)^{\text{morse}(o)} = \chi^{S^1}(L_\beta X),$$

(note $\text{mult}(o) = 1$ by the indivisible condition.)

The proof of the above is based on S^1 -equivariant Morse theory. For divisible classes, the homology theoretic interpretation of the F -count would require more than ordinary S^1 -equivariant Morse homology for the g -energy functional. One would want a g -invariant up to quasi-isomorphism “orbifold Morse homology chain complex” for the quotient stack $[LX/S^1]$, together with (assuming certain finiteness) an algebraically derived $\mathbb{Q}$ -valued characteristic¹ coinciding with the F -count. This is likely related to Wall-Euler characteristic theory in group cohomology theory, which also appears in the more basic Conjecture 1. Existence of this orbifold Morse homology is at the moment unknown; for a finite dimensional candidate see [19].

For a compact X , we say a metric g is taut if the space of class β closed g -geodesics is compact for every β . (With the topology induced by the C^1 topology on the loop space.) It is said to be β -taut if the same holds for just β -class geodesics. A homotopy $\{g_t\}_{t \in [0,1]}$ is said to be β -taut if the space of pairs (o, t) with o class β g_t -geodesic is compact. An example of a metric g , β -taut homotopic to the flat metric is in Figure 1. We can construct such examples which are arbitrarily “far” from being flat. Figure 1 is also an example for the following.

In the other direction, β -tautness fails easily.

Example 1. Let g agree with a flat metric outside a small embedded disk in T^2 , inside which it is rotationally symmetric with a “mushroom” profile: a bulb over a neck, the circumference having a strict local minimum at a waist circle γ_0 , which is then a closed geodesic. Let $S \subset T^2$ be the complement of the open bulb: a compact surface with geodesic boundary γ_0 , with $\pi_1(S)$ free on x, y and $[x, y]$ the boundary class, killed by the inclusion into T^2 . For $k \in \mathbb{N}$ the elements $x[x, y]^k$ represent pairwise distinct nontrivial conjugacy classes of $\pi_1(S)$ (their cyclically reduced lengths differ), all mapping to $\beta := [x]$ in $\pi_1(T^2)$. As ∂S is geodesic, minimizing length in each of these classes produces closed g -geodesics o_k of class β in T^2 ; and since for each L only finitely many free homotopy classes of the compact S contain a loop of length $\leq L$, $\ell_g(o_k) \rightarrow \infty$. So g is not β -taut, and the argument applies to every nontrivial β . The disk may be taken arbitrarily small, so the failure is not curable by localizing the perturbation: the C^{k_0} -closeness in Proposition 9.1 ahead is essential.

Using geodesic confinement based on KAM theory, and the product formula of Theorem 7.1 we show in Section 9 the following:

Theorem 1.3. *Any Finsler metric g on T^2 sufficiently C^k -close to a given flat metric, for a universal k (see Proposition 9.1), is taut homotopic to the flat metric.*

In fact, near the flat metrics the invariant vanishes identically.

¹I am told that in this context a better term is von Neumann dimension.

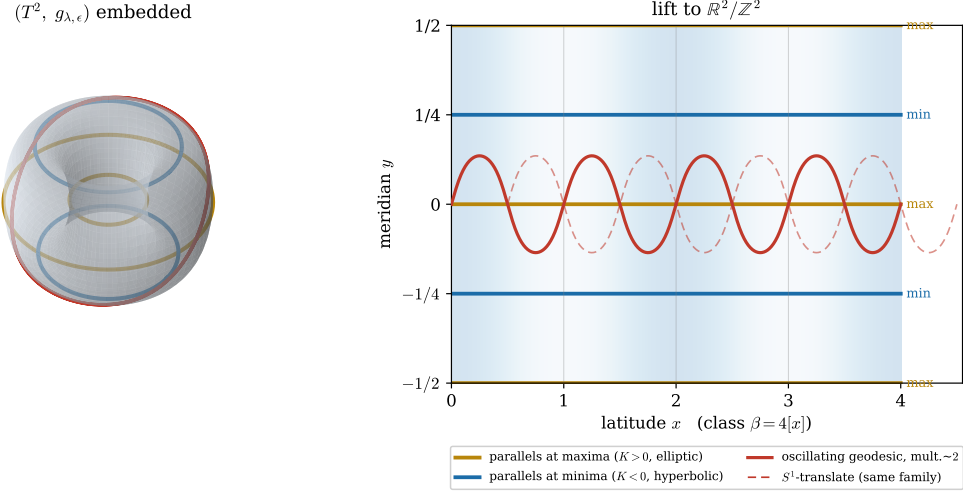


FIGURE 1. Schematic of a β -taut torus metric with mixed geodesic multiplicities in the class $\beta = 4[x]$. In coordinates

$$T^2 = \mathbb{R}^2/\mathbb{Z}^2, \quad g_{\lambda,\epsilon} = dy^2 + \lambda^2 f_\epsilon(y)^2 dx^2, \quad f_\epsilon(y) = 1 + \epsilon \cos(4\pi y),$$

the x -direction is drawn as the latitude direction. The four critical levels of f_ϵ ,

$$y = 0, \quad \frac{1}{4}, \quad \frac{1}{2}, \quad \frac{3}{4},$$

give four geodesics; their 4-covers are class β geodesic strings of multiplicity 4. For $\lambda > (4\sqrt{\epsilon(1+\epsilon)})^{-1}$ there are, in addition, two S^1 -families (S^1 here is acting by rotation in the x -direction) of non-horizontal oscillating geodesic strings in the same class β , one about each maximum of f_ϵ , each of multiplicity 2. This is in part classical Clairaut theory, see for instance [2], [20, §4-4, Example 5]. After a small generic perturbation each family breaks into isolated strings cancelling in the F -count. There are also multiplicity-one oscillating β -class geodesics (one oscillation while winding four times), already for $\lambda > (8\sqrt{\epsilon(1+\epsilon)})^{-1}$; further multiplicity-one families appear as λ increases. In particular the four parallels illustrate the balance $s_1 = 0$ of Proposition 1.5: those at the maxima of f_ϵ have positive curvature along themselves, hence are elliptic, and those at the minima have negative curvature, hence are positive hyperbolic. To see the β -taut homotopy to the flat metric take

$$g_s = dy^2 + \lambda^2 (1 + s\epsilon \cos(4\pi y))^2 dx^2, \quad s : 1 \rightarrow 0.$$

Each g_s is Clairaut-integrable and tautness is then established by explicit tracking of geodesics.

Theorem 1.4 (Proof in Section 10). *There is a universal k such that for every Riemann-Finsler metric g on T^2 sufficiently C^k -close to a flat metric, and every nontrivial class β , g is β -taut and*

$$F(g, \beta) = 0.$$

This vanishing has a complete arithmetic expression in terms of primitive closed geodesics. A Finsler metric that is β -regular for all β is C^∞ -generic by the bumpy metric theorem, in the Riemannian case due to Abraham [1] and Anosov [4] and in the Finsler case to Rademacher-Taimanov [45]; for such g , in each taut class the geodesic strings are isolated points of a compact set, hence finite in number. Fix an indivisible class β_0 , and for $e \geq 1$ let P_e denote the set of primitive closed g -geodesic strings of

class $e\beta_0$, let

$$s_e := \sum_{o \in P_e} i(o) \in \mathbb{Z}$$

be their signed count, where $i(o) := -\text{sign det}(I - \mathcal{R}_o) \in \{\pm 1\}$, with $\mathcal{R}_o$ the linearized return map of o . This is the local Fuller index of o in the normalization (A.2) of Appendix A, under which $i(o) = (-1)^{\text{morse}(o)}$, so that a length minimizing geodesic contributes $+1$. Let b_e be the number of elements of P_e of inverse-hyperbolic type ($\mathcal{R}_o$ has negative eigenvalues).

Proposition 1.5 (Proof in Section 10). *Let g be a Riemann-Finsler metric on T^2 , and β_0 an indivisible class such that g is $d\beta_0$ -taut, $d\beta_0$ -regular, and $F(g, d\beta_0) = 0$ for every $d \geq 1$ — for example any $d\beta_0$ -regular metric C^{k_0} -close to a flat metric, by Theorem 1.4. Then*

$$(1.6) \quad s_e = -b_{e/2} \quad \text{for every } e \geq 1,$$

where $b_{e/2} := 0$ for e odd. Conversely, (1.6) implies $F(g, d\beta_0) = 0$ for all $d \geq 1$.

Corollary 1.7 (Immediate from (1.6)). *In the setting of Proposition 1.5: for e odd the primitive closed geodesics in class $e\beta_0$ come in balancing pairs — if there is one, there is another. If $b_n \geq 1$, that is, if class $n\beta_0$ contains an inverse-hyperbolic primitive closed geodesic, then class $2n\beta_0$ contains at least b_n primitive closed geodesics — a period doubling phenomenon.*

The inverse-hyperbolic type is realized already on T^2 , here is an example.

Example 2. In Fermi form $g = dy^2 + A(x, y)^2 dx^2$ on $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, the curve $\{y = 0\}$ is a unit speed closed geodesic of length 1 as soon as $A(x, 0) \equiv 1$ and $\partial_y A(x, 0) \equiv 0$; its Gauss curvature along itself is then $K(x) = -\partial_y^2 A(x, 0)$, and its normal variational equation is the Hill equation $J'' + KJ = 0$. Taking

$$A(x, y) = 1 - \frac{1}{2} \left(1 + \frac{5}{\pi^2} \cos 2\pi x \right) \sin^2 \pi y$$

gives a metric on T^2 , with $A > 0$ and $K(x) = \pi^2 + 5 \cos 2\pi x$ along the geodesic $o = \{y = 0\}$, whose linearized return map $\mathcal{R}_o$ has $\text{tr } \mathcal{R}_o < -2$ (a numerical evaluation of the Hill discriminant gives $-2.158\dots$). In particular o is inverse-hyperbolic, and $\det(I - \mathcal{R}_o) > 0$ while $\det(I - \mathcal{R}_o^2) < 0$, so $i(o) = -1$ while $i(o^2) = +1$: the contribution flips on doubling, consistently with the conjugate point exhibited below. The parameters are dictated by Hill theory, the inverse-hyperbolic locus being the antiperiodic instability tongue, which emanates from the constant value $K \equiv \pi^2$ and is opened by the first harmonic of K .

Note that if o is inverse-hyperbolic, some normal Jacobi field J along o satisfies $J(s + \ell_g(o)) = -\lambda J(s)$ and hence vanishes somewhere: o carries a conjugate point, and Sturm comparison gives $\ell_g(o) \geq \pi/\sqrt{\max_o K}$. So under the hypotheses of Theorem 1.4 the curvature is uniformly small, and inverse-hyperbolic primitive geodesics occur only in the classes $e\beta_0$ with e at least of the order of $(\sup |K|)^{-1/2}$; the period doubling constraint of Proposition 1.5 is in effect a statement about high classes, in contrast to odd class case where the constraint affects all classes.

Remark 1.8. There is nothing particularly special about T^2 except that we can very concretely produce a whole universe of taut metrics on it (e.g. near flat, Clairaut-integrable type, etc.). There are analogous arithmetic laws for the primitive geodesic strings of a metric on a closed manifold which, for an indivisible class β , is $d\beta$ -taut and $d\beta$ -regular for all $d \geq 1$. The source is dimension free: exactly as in the proof of Proposition 1.5, the covering bijection gives, for every d , the identity between integers

$$dF(g, d\beta) = \sum_{e|d} e \sum_{o \in P_e} i(o^{d/e}),$$

with P_e the set of primitive strings of class $e\beta$ and i the local Fuller index (A.2), so that the values of F along the ray constrain the signed iterate counts of the primitive strings. The sign patterns of $i(o^k)$

in k are governed by the eigenvalue configuration of $\mathcal{R}_o$; on T^2 , Lemma 10.1 reduces them to the pair (s_e, b_e) , and the vanishing of F gives (1.6). In general the resulting arithmetic is more sophisticated, and to avoid overloading the paper we do not present it. If F is a topological quantity, as conjectured, computing its values along the ray may be easier than it looks.

Theorem 1.3 settles the following question near the flat metric.

Question 1. Is every β -taut metric on T^2 β -taut homotopic to the flat metric?

Proposition 1.5 is not accessible to S^1 -equivariant Morse homology techniques, as the latter doesn't suitably track isotropy information. Nevertheless, there is a family resemblance with the resonance identities of closed geodesic and closed characteristic theory, where finite collections of periodic objects satisfy identities involving average Euler characteristics and mean indices; see, for example, [44, 27] and the later noncontractible resonance identities [30]. The constraints of the present paper are of a different character: the resonance identities are real valued relations driven by the asymptotics of the index, while ours are exact integer identities among the signed counts of primitive closed geodesics, driven by the vanishing of a rational count.

Remark 1.9. The near flat statements extend to Reeb flows on the unit cotangent bundle. Let g_0 be a flat metric on T^2 , let $C = S_{g_0}^* T^2$ with its canonical contact form λ_{g_0} , and let R be the Reeb vector field of a contact form $f\lambda_{g_0}$ on C with $\|f - 1\|_{C^k}$ sufficiently small, for a universal k . Then, in the flat trivialization $C \cong T^2 \times S^1$, every closed R -orbit has the free homotopy class $(\beta, 0)$ of a lifted flat geodesic, for some $\beta \neq 0$, with period close to $|\beta|$; in particular R has no contractible closed orbits. Moreover R is taut homotopic, through Reeb vector fields, to the flat geodesic Reeb field $R^{\lambda_{g_0}}$, $F(R, (\beta, 0)) = 0$ for every nontrivial β , and when the closed R -orbits in the classes $(e\beta_0, 0)$, $e \geq 1$, are nondegenerate, their signed primitive counts obey the law of Proposition 1.5. All of this follows from the results above through the correspondence $f\lambda_{g_0} = \lambda_{g_f}$ between fiberwise convex contact forms and Finsler metrics, together with the lift of Section 5.1, which matches orbit strings with geodesic strings preserving classes, multiplicities and nondegeneracy. As the verification is routine but lengthy, we omit the details to keep the paper focused.

We will show that the count $F(g, \beta) \in \mathbb{Q}$ can be defined for any β -taut g (Section 5). What follows is a special version of the Conjecture 2 (appearing further ahead) and can be understood as a divisible analogue of Corollary 1.2 in the aspherical case.

Conjecture 1 (Aspherical topological formula for F). *Let X be a closed aspherical manifold, and let β be a nontrivial free homotopy class. Suppose that X admits a β -taut metric. Then the Borel construction*

$$(L_\beta X)_{hS^1} := L_\beta X \times_{S^1} ES^1$$

is Wall-Euler finite, and for every β -taut metric g ,

$$F(g, \beta) = \chi_{\text{Wall}}((L_\beta X)_{hS^1}).$$

Equivalently, set $G = \pi_1(X)$, let $C_G(\gamma_\beta)$ be the centralizer of γ_β in G and $Q_\beta = C_G(\gamma_\beta)/\langle \gamma_\beta \rangle$. where $\beta = [\gamma_\beta]$ is the conjugacy class of $\gamma_\beta \in G$. Then Q_β is Wall-Euler finite and

$$F(g, \beta) = \chi_{\text{Wall}}(Q_\beta).$$

The conjecture implies that $F(g, \beta)$ is independent of the β -taut metric g , for closed aspherical manifolds. For background on rational Euler characteristics of groups, we refer to Wall's original note [49] and to Brown's treatment of Euler characteristics of groups in [17, Chapter IX]; see also Brown [16] for Euler characteristics of discrete groups and G -spaces.

Note that if we assume the above conjecture, then $F(g, \beta) = 0$ for every β -taut metric g on T^n , so that the conclusions of Proposition 1.5 hold for any such g which is moreover $d\beta_0$ -regular for all d , without a near flat hypothesis.

The following is a rather important open question in our context:

Question 2. Suppose that X is closed, if X admits a β -taut metric for some nontrivial β , must X be aspherical?

Remark 1.10. A possible route to a counterexample for Question 2 is a mapping torus of a non-aspherical Z by a diffeomorphism with “hyperbolic behavior”. In the other direction, asphericity is not sufficient: by results of Mess [35] and Belegradek [9, Corollary 5.1], in every dimension $n \geq 4$ there are closed aspherical n -manifolds whose fundamental groups contain nontrivial infinitely divisible elements, and hence, by Proposition 1.11 below, classes β for which no β -taut metric exists.

Proposition 1.11. *Let X be a closed manifold and let $\beta \neq 1$ be a free homotopy class which is infinitely divisible in $\pi_1(X)$: for arbitrarily large $n \in \mathbb{N}$, a representative of β in $\pi_1(X)$ is an n -th power. Then X admits no β -taut metric.*

Proof. Take any metric g and let $l > 0$ be the systole of g , that is

$$l = \inf_{\{o \mid o \text{ is noncontractible}\}} \ell_g(o).$$

For n as in the statement, write a representative of β as h_n^n , with h_n necessarily noncontractible since $\beta \neq 1$. A length-minimizing closed geodesic in the free homotopy class of h_n exists as X is closed, and its n -fold cover is a class β closed g -geodesic of length at least $n \cdot l$. So the class β closed geodesics of g have unbounded length and g is not β -taut. $\square$

1.1. Geodesics in mapping tori and negative sectional curvature. A theorem of Preissmann [43] says that there are no negative sectional curvature metrics on compact products. For fibrations, the most obvious analogue of Preissmann’s theorem fails even assuming compactness. By Thurston’s fibered hyperbolization theorem, a closed surface bundle over S^1 with fiber of negative Euler characteristic is hyperbolic if and only if its monodromy is pseudo-Anosov; see [48, 47]. What follows is a kind of partial, “only if” Finsler analogue in higher dimensions, noting that an isometry of a manifold has “anti-Anosov behavior” (for some reasonable generalization of pseudo-Anosov to higher dimensions).

We denote by $\pi_1^{inc}(X)$ the set of free homotopy classes of loops incompressible to the ends, in the sense of Definition 2.1, (by convention nonconstant classes when X is closed).

Corollary 1.12 (Corollary of Corollary 2.27). *Let X_ϕ be the mapping torus of an isometry $\phi : Z \rightarrow Z$ of a complete taut Riemannian metric on Z , with Z finite type, and where $\beta_Z \in \pi_1^{inc}(Z)$ is indivisible. Let $\beta \in \pi_1^{inc}(X_\phi)$ denote the image of β_Z under inclusion. Then:*

- (1) $\chi^{S^1}(L_\beta X_\phi) = 0$.
- (2) X_ϕ does not admit a forward complete, β -regular Riemann-Finsler metric with an odd number of class β geodesic strings.
- (3) X_ϕ does not admit a complete negative sectional curvature Riemannian metric.

Example 3. The corollary applies to the mapping torus of a finite type surface of genus at least one (the genus condition allows us to find a nontrivial class in $\pi_1^{inc}(X_\phi)$, we also need a suitable taut metric e.g. a nonpositive curvature metric, with an isometry). The conclusion that X_ϕ does not admit

a complete negative sectional curvature metric follows from Thurston's classification². However the arithmetic constraints on geodesic counts seem interesting even for a surface map.

If we further assume that Z is aspherical, the first part of Corollary 1.12 can be proved via group cohomology theory, for which we already provided some references. But the main point is that in our setup the theorem has an elementary proof, once the general product formula of Theorem 7.1 for the F -count is established. The proof of the latter does involve a tricky perturbation, but Fuller index theory allows us to do this abstractly, perturbing a vector field rather than the metric/energy functional.

Organization of the paper. Section 2 sets up geodesic strings, taut metrics and taut deformations, states the supporting theorems, and contains the applications to sky catastrophes. Section 3 proves the compactness results underlying tautness, in particular Theorem 2.5. Section 4 collects Reeb-flow preliminaries, and Section 5 constructs the invariant F and proves its taut deformation invariance. Section 6 proves the Euler characteristic formula for indivisible classes, and Sections 7 and 8 prove the product and fibration formulas. Section 9 proves the near flat rigidity Proposition 9.1 by KAM confinement, and Section 10 proves the vanishing Theorem 1.4 and the arithmetic law of Proposition 1.5. Appendix A develops the Fuller index and its continuation theory in the noncompact setting, Appendix B performs a local fibered computation of F used in the product formula, and Appendix C contains the facts about sky catastrophes used in Section 2.0.2.

2. SETUP AND SUPPORTING THEOREMS

Proofs of many results here are postponed till subsequent sections.

Terminology 1. From now on, all our metrics are Riemann-Finsler (a.k.a. Finsler) metrics unless specified to be Riemannian, and usually denoted by just g . Completeness, always means forward completeness, and it is an assumption for all our metrics. Curvature always means sectional curvature in the Riemannian case and flag curvature in the Finsler case. Thus we will usually just say complete metric g , for a forward complete Riemann-Finsler metric. A reader may certainly choose to interpret all metrics as Riemannian metrics, completeness as standard completeness, and curvature as sectional curvature.

In what follows, $\pi_k(X)$ denotes the usual based homotopy group, with basepoints suppressed when no confusion can arise. We do not use $\pi_1(X)$ to denote free homotopy classes.

Definition 2.1. Let X be a smooth manifold. Fix an exhaustion by nested compact sets $\bigcup_{i \in \mathbb{N}} K_i = X$, $K_i \supset K_{i-1}^\circ$ for all $i \geq 1$. A free homotopy class β of loops in X is called **end compressible** if, for every i , it has a representative contained in $X - K_i$. We say that β is **end incompressible** (or incompressible to the ends) if it is not end compressible.

Let $\pi_1^{inc}(X)$ denote the set of such end incompressible free homotopy classes. When X is compact, we set $\pi_1^{inc}(X)$ to be the set of nonconstant free homotopy classes of loops in X .

It is easily seen that the above is well defined (independent of the choice of an exhaustion) and moreover any homeomorphism $X_1 \rightarrow X_2$ of a pair of manifolds induces a set isomorphism $\pi_1^{inc}(X_1) \rightarrow \pi_1^{inc}(X_2)$.

Denote by $L_\beta X$ the class $\beta \in \pi_1^{inc}(X)$ component of the free loop space of X , with its compact open topology. Let g be a complete metric on X , and let $S(g, \beta) \subset L_\beta X$ denote the subspace of all constant speed parametrized, smooth, closed g -geodesics in class β .

²Suitably adjusted, one may also need finite geometry (volume) assumptions.

Definition 2.2. We say that a metric g on X is β -**taut** if it is complete and $S(g, \beta)$ is compact. We will say that g is **taut** if it is β -taut for each $\beta \in \pi_1^{inc}(X)$.

Lemma 2.3. *A complete metric g with nonpositive curvature satisfies:*

- *All of its closed geodesics are minimizing in their free homotopy class.*
- *It is taut.*

Proof. The first part is a standard consequence of the Cartan-Hadamard theorem. The second part follows by the first part and Corollary 3.3. $\square$

Definition 2.4. Let $\beta \in \pi_1^{inc}(X)$, and let g_0, g_1 be a pair of β -taut metrics on X . A β -**taut deformation** between g_0, g_1 , is a continuous (in the topology of C^0 convergence on compact sets) family $\{g_t\}$, $t \in [0, 1]$ of complete metrics on X , such that

$$S(\{g_t\}, \beta) := \{(o, t) \in L_\beta X \times [0, 1] \mid o \in S(g_t, \beta)\}$$

is compact. We say that $\{g_t\}$ is a **taut deformation** if it is β -taut for each $\beta \in \pi_1^{inc}(X)$. The above definitions of tautness are extended naturally to the case of a smooth fibration $X \hookrightarrow P \rightarrow [0, 1]$, with a smooth fiber-wise family of metrics.

A useful criterion for β -tautness is the following.

Theorem 2.5 (Proof in Section 3.1). *Let $\beta \in \pi_1^{inc}(X)$. Let $\{g_t\}_{t \in [0, 1]}$ be a continuous family of complete metrics on X . Suppose that:*

$$\sup_t \left| \sup_{o \in S(g_t, \beta)} \ell_{g_t}(o) - \inf_{o \in S(g_t, \beta)} \ell_{g_t}(o) \right| < \infty,$$

where ℓ_{g_t} is the length functional with respect to g_t , then $\{g_t\}$ is β -taut.

For example, the hypothesis is trivially satisfied if g_t have the property that all their class β closed geodesics are minimizing in their homotopy class.

Corollary 2.6. *If g_t , $t \in [0, 1]$ have nonpositive curvature then $\{g_t\}$ is taut.*

Proof. This follows by Theorem 2.5 and by Lemma 2.3. $\square$

2.0.1. *The geodesic string counting invariant F .* Let $\mathcal{G}(X)$ be the set of equivalence classes of taut metrics g , where g_0 is equivalent to g_1 whenever there is a taut deformation between them. We may denote an equivalence class by its representative g by a slight abuse of notation. The following is proved in Section 5.

Theorem 2.7. *For each manifold X there is a natural, nontrivial functional:*

$$F : \mathcal{G}(X) \times \pi_1^{inc}(X) \rightarrow \mathbb{Q},$$

extending the definition in (1.1).

For g β -regular and β -taut we have already defined $F(g, \beta)$ at the beginning of the Introduction. For a general β -taut g we define $F(g, \beta)$ in terms of the Fuller index.

Corollary 2.8. *Suppose for a pair g_0, g_1 of β -taut metrics on X :*

$$F(g_0, \beta) \neq F(g_1, \beta),$$

then any path $\{g_t\}$, connecting g_0, g_1 , is not β -taut.

Proof. The fact that any connecting $\{g_t\}$ is not β -taut is just a direct corollary of the theorem above. $\square$

I was unable to find such a pair g_0, g_1 , in fact there is strong evidence to believe the following:

Conjecture 2. *F does not depend on the choice of a smooth structure and taut metric. In other words we have the following. Let X be a topological manifold, define:*

$$\mathcal{S}(X) : \pi_1^{\text{inc}}(X) \rightarrow \mathbb{Q} \sqcup \{\infty\}$$

by:

$$\mathcal{S}(X)(\beta) = \begin{cases} \infty, & \text{if } X \text{ does not admit a smooth structure and a } \beta\text{-taut Finsler metric.} \\ F(g, \beta), & \text{if } g \text{ is a } \beta\text{-taut Finsler metric on } X. \end{cases},$$

then $\mathcal{S}$ is well-defined and hence determines a topological invariant of topological manifolds. So if $f : X_1 \rightarrow X_2$ is a homeomorphism then $\mathcal{S}(X_1)(\beta) = \mathcal{S}(X_2)(f_*(\beta))$.

Theorem 2.16 in the following section partially proves this conjecture.

Remark 2.9. The smooth manifold version of this conjecture, i.e. that $\mathcal{S}(X)$ is a smooth manifold invariant, is implied by the Reeb sky catastrophe conjecture described in the next section. However, the above seems to be much more basic as will be apparent from the proof of Theorem 2.16.

2.0.2. Applications to sky catastrophes.

Definition 2.10 (Preliminary). A **sky catastrophe** for a smooth family $\{X_t\}$, $t \in [0, 1]$, of nonvanishing vector fields on a closed manifold M is a continuous family of closed orbit strings $\tau \mapsto o_{t_\tau}$, o_{t_τ} is an orbit string of X_{t_τ} , $\tau \in [0, \infty)$, such that the period of o_{t_τ} is unbounded from above.

Sky catastrophes first appeared in the work of Fuller [22]. One general definition appears in [46], which we review in the Appendix C, the main point of this definition is that there are no regularity assumptions on $\{X_t\}$. The preliminary definition becomes equivalent to the more general definition given certain regularity conditions on the family $\{X_t\}$.

The following corollary of Theorem 2.5 may be of independent interest.

Corollary 2.11. *Sky catastrophes of vector fields on closed manifolds are not geodesible by metrics all of whose geodesics are minimizing. That is, if a family $\{X_t\}$ has a sky catastrophe, there is no family $\{g_t\}$ of metrics (all of whose geodesics are minimizing in their homotopy class) such that the orbits of X_t are unit speed parametrized g_t -geodesics.*

Fuller at the end of [23] has asked for any metric conditions on vector fields to rule out sky catastrophes. By the above, nonpositivity of curvature is one such condition. So this is a partial answer to his question.

The main conjecture in [46] is that geodesible, and more generally Reeb sky catastrophes ($\{X_t\}$ are Reeb) are unstable, in the sense that there is a C^0 nearby family $\{X'_t\}$ which does not have a sky catastrophe (at least in the fixed class β). Let's call this **Reeb sky catastrophe conjecture**. The reason to conjecture such a thing, is that a theorem in [46] says that if $\{o_{t_\tau}\}_\tau$ is a sky catastrophe for a family of Reeb vector fields $\{X_t\}$ then the length of the corresponding projection $\tau \mapsto t_\tau$ is infinite. (Assuming some minor regularity on $\{o_{t_\tau}\}$.) The latter suggests that the structure of such a sky catastrophe is very fragile.

Remark 2.12. The Reeb sky catastrophe conjecture implies the geodesible Seifert conjecture (apparently still an open problem), by the main result of [46]. Hence, this is a subtle question.

The following is a stronger form of Corollary 2.8.

Theorem 2.13. *Suppose for a pair g_0, g_1 of β -taut metrics on a closed manifold M :*

$$F(g_0, \beta) \neq F(g_1, \beta),$$

then any path $\{g_t\}$, connecting g_0, g_1 , has a sky catastrophe, meaning that the corresponding family $\{X_t\}$ of Reeb vector fields on the unit cotangent bundle (cf. Section 5) has a sky catastrophe as in Definition C.1.

Proof. This directly follows by [46, Theorem 3.2]. $\square$

So that if such a pair g_0, g_1 exists, the Reeb sky catastrophe conjecture is disproved and in a rather extreme fashion.

2.0.3. Basic results on the invariant F .

Definition 2.14. Let β be a free homotopy class of loops in X . Choose a representative of β and a base point x_0 on its image. This choice determines an element

$$\beta_{x_0} \in \pi_1(X, x_0)$$

well-defined up to inner automorphism.

- An element $\gamma \in \pi_1(X, x_0)$ is **indivisible** if whenever $\gamma = \alpha^k$ for some $\alpha \in \pi_1(X, x_0)$ and $k > 0$, then $k = 1$.
- An element $\gamma \in \pi_1(X, x_0)$ is a **k -power** if $\gamma = \alpha^k$, $k > 1$, for some α which is not an n -power for any $n > 1$.
- An element $\gamma \in \pi_1(X, x_0)$ is **atomic** if $\gamma = \alpha^k$, for some $k \geq 1$, with α indivisible.

We say that the free homotopy class β is indivisible, respectively a k -power, respectively atomic, if β_{x_0} has the corresponding property for one, equivalently any, such choice of x_0 and representative. This is well-defined because the corresponding notions are invariant under inner automorphism. Equivalently, β is a k -power precisely when it admits a smooth representative with multiplicity k whose underlying simple representative is indivisible.

Note that a class $\beta \in \pi_1^{inc}(X)$ is indivisible iff any representative of this class is not multiply covered.

Example 4. Let g be a Riemannian metric with negative sectional curvature on a closed manifold X and $\beta \in \pi_1^{inc}(X)$ a class represented by a multiplicity n closed geodesic, then

$$(2.15) \quad F(g, \beta) = \frac{1}{n}.$$

In particular, if β is indivisible then $F(g, \beta) = 1$. More generally, (2.15) holds whenever g has a unique and nondegenerate geodesic string in class β , where nondegenerate is as in the Introduction.

If $\beta \in \pi_1^{inc}(X)$ is indivisible, then it is easy to see that the reparametrization S^1 action on $L_\beta X$ is free (Section 6), so that $H_*^{S^1}(L_\beta X, \mathbb{Z}) \simeq H_*(L_\beta X/S^1, \mathbb{Z})$, where $H_*^{S^1}(L_\beta X, \mathbb{Z})$ denotes the S^1 -equivariant homology. Moreover, we have the following result proved in Section 6.

Theorem 2.16. *Suppose that $\beta \in \pi_1^{inc}(X)$ is indivisible, and X admits a β -taut metric. Then $H_*^{S^1}(L_\beta X, \mathbb{Z})$ has finite total rank. Denote by $\chi^{S^1}(L_\beta X)$ the Euler characteristic of this homology. Then for any β -taut metric g on X :*

$$F(g, \beta) = \chi^{S^1}(L_\beta X).$$

Consequently, $F(g, \beta)$ is independent of the choice of β -taut metric g , for indivisible β . Furthermore, Conjecture 2 holds on the subset of classes β which are indivisible.

At least for surfaces β -taut metrics (distinct from nonpositive curvature metrics) can be constructed explicitly, recall the example of Figure 1.

We can use the above, and the product formula of Theorem 7.1 to get:

Theorem 2.17 (Proof in Section 8). *Every rational number has the form $F(g, \beta)$ for some β -taut Riemannian g on some compact manifold X and for some β .*

2.0.4. Applications to existence of negative curvature metrics. We are going to give a certain generalization of Preissmann's theorem to fibrations, with possibly noncompact fibers, also replacing the negative sectional curvature condition by a significantly weaker condition. Though the definitions may at first seem hard to understand, we just want enough generality to incorporate interesting “flat bundle” examples (with anti-Anosov or periodic holonomy behavior). The latter are of interest in 3-manifold theory, for instance vis-à-vis Thurston's classification [47].

Definition 2.18. For a fiber bundle $Z \hookrightarrow X \rightarrow Y$, we say that $\beta \in \pi_1^{\text{inc}}(X)$ is a **fiber class** if it lies in the image of the set map

$$i_Z : \pi_1^{\text{inc}}(Z) \rightarrow \pi_1^{\text{inc}}(X)$$

induced by inclusion of a fiber.

Lemma 2.19. *Let*

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a smooth fiber bundle with connected fiber and connected base. Assume

$$\pi_2(Y) = 0.$$

Fix $y_0 \in Y$, and let

$$H = \pi_1(Y, y_0)$$

act on $\pi_1^{\text{inc}}(Z_{y_0})$ by holonomy. Let $\beta \in \pi_1^{\text{inc}}(X)$ be a fiber class, and define

$$\mathcal{R}_{y_0}(\beta) = \{ \alpha \in \pi_1^{\text{inc}}(Z_{y_0}) \mid (i_{y_0})_*(\alpha) = \beta \}.$$

Then $\mathcal{R}_{y_0}(\beta)$ is a single H -orbit. In particular, if

$$\beta_Z, \beta'_Z \in \mathcal{R}_{y_0}(\beta),$$

then

$$H \cdot \beta_Z = H \cdot \beta'_Z.$$

Thus the holonomy orbit associated to the fiber class β is independent of the choice of fiber representative β_Z .

Proof. Choose basepoints $z_0 \in Z_{y_0}$ and $x_0 = i_{y_0}(z_0)$. Since $\pi_2(Y) = 0$, the homotopy long exact sequence gives an exact sequence of based groups

$$1 \rightarrow \pi_1(Z_{y_0}, z_0) \xrightarrow{(i_{y_0})_*} \pi_1(X, x_0) \xrightarrow{p_*} \pi_1(Y, y_0) \rightarrow 1.$$

Thus we identify $\pi_1(Z_{y_0}, z_0)$ with a normal subgroup of $\pi_1(X, x_0)$, and the quotient is $\pi_1(Y, y_0)$.

Let $\alpha, \alpha' \in \mathcal{R}_{y_0}(\beta)$. Choose based representatives of these fiber free homotopy classes. Since they determine the same total free homotopy class β in X , the corresponding elements of $\pi_1(X, x_0)$ are conjugate. Thus, after conjugating inside $\pi_1(Z_{y_0}, z_0)$ if necessary, there exists $x \in \pi_1(X, x_0)$ such that

$$\alpha' = x\alpha x^{-1}.$$

Let

$$h = p_*(x) \in H.$$

Conjugation by x on the normal subgroup $\pi_1(Z_{y_0}, z_0)$ induces the holonomy action of h , up to an inner automorphism of $\pi_1(Z_{y_0}, z_0)$. Passing to free homotopy classes in the fiber, we get

$$\alpha' = h \cdot \alpha.$$

Thus all fiber representatives of the total free homotopy class β belong to a single holonomy orbit.

Conversely, if $\alpha' = h \cdot \alpha$, choose $x \in \pi_1(X, x_0)$ with $p_*(x) = h$. Then $x\alpha x^{-1}$ represents the same total free homotopy class in X as α , and its fiber free homotopy class is $h \cdot \alpha$. Hence the holonomy orbit is exactly the set of fiber representatives of β . $\square$

We assume from now on that Z, Y are connected. For a fiber class $\beta \in \pi_1^{\text{inc}}(X)$, set

$$\mathcal{R}(\beta) = \{\alpha \in \pi_1^{\text{inc}}(Z) \mid (i_Z)_*(\alpha) = \beta\}.$$

Assume that $\pi_2(Y) = 0$, and put

$$H = \pi_1(Y, y_0).$$

By Lemma 2.19, the holonomy orbit

$$S = H \cdot \beta_Z$$

is independent of the choice of the fiber representative $\beta_Z \in \mathcal{R}(\beta)$. If another fiber inclusion is chosen, the corresponding holonomy actions are abstractly conjugate.

Definition 2.20. Let

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a smooth fiber bundle, with Y and $\pi_2(Y) = 0$, and let $\beta \in \pi_1^{\text{inc}}(X)$ be a fiber class. We say that p is **β -holonomy-finite** if for one, equivalently any, $\beta_Z \in \mathcal{R}(\beta)$, the orbit

$$S = \pi_1(Y, y_0) \cdot \beta_Z$$

is finite.

Definition 2.21. Let

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a smooth fiber bundle, and let g be a Riemannian metric on X . Let

$$T^{\text{vert}}X = \ker(dp)$$

and let

$$\pi^{\text{vert}} : TX \rightarrow T^{\text{vert}}X$$

be the g -orthogonal projection. We say that p is **g -parallel** if

$$\nabla \pi^{\text{vert}} = 0.$$

Equivalently,

$$\nabla_A(\pi^{\text{vert}}B) = \pi^{\text{vert}}(\nabla_A B)$$

for all vector fields A, B on X .

Definition 2.22. Let $\beta \in \pi_1^{\text{inc}}(X)$ be a fiber class, for a smooth fibration

$$p : Z \hookrightarrow (X, g) \rightarrow (Y, g_Y).$$

We say that this is a **β -periodic fibration** if:

- (1) g is a complete Riemannian metric.
- (2) Z, Y are connected and $\pi_2(Y) = 0$.
- (3) p is g -parallel and is a Riemannian submersion.

- (4) p is β -holonomy-finite.
- (5) the fiber metrics g_y on Z_y are taut.

The metrics g, g_Y may be kept implicit. In the above definition of a β -periodic fibration and the hypothesis of the following theorem we need the auxiliary metric g on X to be Riemannian, and there is no obvious extension of the theorem to the Riemann-Finsler case. However, the conclusions of the theorem are for Riemann-Finsler metrics. The following is perhaps the main technical result of the paper, proved in Section 8.

Theorem 2.23. *Let $p : Z \hookrightarrow (X, g) \rightarrow (Y, g_Y)$ be a β -periodic fibration, where $\beta \in \pi_1^{\text{inc}}(X)$ is a fiber class. Suppose further that Y is connected, closed, $\chi(Y) \neq \pm 1$ and is such that all smooth closed contractible g_Y -geodesics in Y are constant. Then the following holds:*

- X does not admit a complete Riemannian metric with negative curvature.
- X does not admit a forward complete Riemann-Finsler metric with a unique nondegenerate class β geodesic string.

Note that $\chi(Y) \neq 1$ is of course essential, as the trivial fibration $X \rightarrow \{pt\}$, with X admitting a complete negatively curved metric, will satisfy the hypothesis. The condition that there is a fiber class $\beta \in \pi_1^{\text{inc}}(X)$ is also essential; as any vector bundle over a manifold admitting a Riemannian metric of negative curvature also admits a metric of negative curvature, Anderson [3].

Corollary 2.24 (Proof in Section 8). *Let Y, Z be smooth manifolds, let $\beta_Z \in \pi_1^{\text{inc}}(Z)$, and assume that $\chi(Y) \neq \pm 1$. Suppose that Y is closed and admits a metric all of whose contractible geodesics are constant, for instance a nonpositive curvature Riemannian metric, and suppose that Z admits a taut metric. Let*

$$X = Y \times Z$$

with projection $X \rightarrow Y$, let $i : Z \rightarrow X$ be a fiber inclusion, and set

$$\beta = i_*(\beta_Z).$$

Then:

- X does not admit a complete Riemannian metric of negative sectional curvature.
- X does not admit a forward complete Riemann-Finsler metric with a unique and nondegenerate class β geodesic string.

Example 5. Take Y any nonpositively curved closed manifold and Z a surface of genus at least one, possibly infinite type.

This theorem is very close to being sharp, for example the conclusion of the corollary is false if $Y = S^1$ and $Z = S^1 \times \mathbb{R}$. As $X = T^2 \times \mathbb{R}$ admits the warped product metric $g_{\mathbb{R}} \times_{e^t} g_{T^2}$ (with respect to the function $f = e^t$ on $\mathbb{R}$) where $g_{T^2}, g_{\mathbb{R}}$ are the flat metrics. This warped product has constant negative sectional curvature -1 . So it is essential that not only $\pi_1(Z)$ be nontrivial but also that there is a class incompressible to the ends. Of course, $\chi(Y) \neq 1$ is also obviously essential, otherwise we may take $Y = pt$. The condition $\chi(Y) \neq -1$ is however not obviously essential.

A further basic set of examples for the theorem is obtained by starting with a Riemannian manifold (Y, g_Y) , and any homomorphism

$$(2.25) \quad \phi : \pi_1(Y, y_0) \rightarrow \text{Isom}(Z, g_Z), \quad (\text{the group of all isometries}).$$

where g_Z is a taut Riemannian metric and $\beta_Z \in \pi_1^{\text{inc}}(Z)$. (For example (Z, g_Z) is a non-simply connected complete hyperbolic surface of genus at least one). Suppose further:

(1) The orbit

$$\text{Orb}(\beta_Z) = \{\phi_*(\gamma)(\beta_Z) \mid \gamma \in \pi_1(Y, y_0)\}$$

is finite.

(2) Y is closed and connected.

(3) All contractible smooth closed g_Y geodesics in Y are constant.

We have the obvious induced diagonal action

$$\begin{aligned} \pi_1(Y, y_0) &\rightarrow \text{Diff}(Z \times \tilde{Y}), \text{ (the group of all diffeomorphisms),} \\ \gamma &\mapsto ((z, y) \mapsto (\phi(\gamma)(z), \gamma \cdot y)), \end{aligned}$$

for $\tilde{Y}$ the universal cover of Y . Taking the quotient of $Z \times \tilde{Y}$ by this action, we get an associated “flat” bundle $Z \hookrightarrow X_\phi \xrightarrow{p} Y$, with a metric g_ϕ induced from the product metric $\tilde{g} = g_Z \oplus g_Y$, on the covering space $q : Z \times \tilde{Y} \rightarrow Z \times Y$. The natural projection $p : X_\phi \rightarrow Y$ is then a Riemannian submersion and is a g_ϕ -parallel. The condition on the finiteness of Orb implies that p is β -holonomy finite where β is the image by $(i_Z)_* : \pi_1^{\text{inc}}(Z) \rightarrow \pi_1^{\text{inc}}(X_\phi)$ of β_Z . It follows that p is a β -periodic fibration. Hence $p : (X_\phi, g_\phi) \rightarrow (Y, g_Y)$ satisfies the hypothesis of Theorem 2.23.

Example 6. Suppose we have $\beta_Z \in \pi_1^{\text{inc}}(Z)$, and let $\phi : Z \rightarrow Z$ be an isometry of a taut metric g_Z , such that Orb is finite. Then for $\beta = (i_Z)_*(\beta_Z)$, by the construction above, the mapping torus

$$(Z, g_Z) \hookrightarrow (X_\phi, g_\phi) \xrightarrow{\pi} S^1$$

has the structure of a β -periodic fibration, satisfying the hypothesis of Theorem 2.23.

The example can be made more concrete as follows.

Lemma 2.26. *Let Z be a complete Riemannian manifold of finite type. Then every isometry of Z has finite orbits on $\pi_1^{\text{inc}}(Z)$.*

Proof. We first prove that, for every $L > 0$, only finitely many classes in $\pi_1^{\text{inc}}(Z)$ have representatives of length at most L . Choose a compact core

$$K_0 \subset Z$$

such that $Z - K_0$ is a finite union of collar ends. Since Z is complete, the closed L -neighborhood

$$K_L = N_L(K_0)$$

is compact. If a representative of an end-incompressible class lies entirely in $Z - K_0$, then it lies in a collar end and can be pushed arbitrarily far out, contradicting end-incompressibility. Therefore every representative of an end-incompressible class must meet K_0 . If such a representative has length at most L , then its image lies in $K_L = N_L(K_0)$.

It remains to use compactness. A compact Riemannian manifold, or compact domain after a harmless thickening of K_L , contains only finitely many free homotopy classes with representatives of length at most L . For instance, choose a finite good cover with Lebesgue number $\delta > 0$. A loop of length at most L is encoded by an edge loop in the nerve of the cover, of combinatorial length bounded by $N(L, \delta)$, and there are only finitely many such combinatorial loops. Hence there are only finitely many end-incompressible classes represented by loops of length at most L .

Now let $\phi : Z \rightarrow Z$ be an isometry and let

$$\beta_Z \in \pi_1^{\text{inc}}(Z).$$

Choose a loop c representing β_Z , and set

$$L = \ell(c).$$

Then $\phi^k(c)$ represents $\phi_*^k(\beta_Z)$ and has the same length L for every $k \in \mathbb{Z}$. Since ϕ preserves end-incompressibility, the orbit

$$\{\phi_*^k(\beta_Z) \mid k \in \mathbb{Z}\}$$

is contained in the finite set of end-incompressible classes represented by loops of length at most L . Therefore the orbit is finite. $\square$

Corollary 2.27. *Let*

$$(Z, g_Z) \hookrightarrow (X_\phi, g_\phi) \rightarrow S^1$$

be the mapping torus of an isometry of Z, g_Z , with Z finite type and g_Z taut. Let $\beta \in \pi_1^{\text{inc}}(X_\phi)$ be the image of some $\beta_Z \in \pi_1^{\text{inc}}(Z)$ under fiber inclusion, as previously.

- (1) X_ϕ does not admit a complete Riemannian metric with negative curvature.
- (2) X_ϕ does not admit a forward complete Riemann-Finsler metric with a unique nondegenerate class β geodesic string.
- (3) If β is indivisible, $\chi^{S^1}(L_\beta X_\phi) = 0$ and so X_ϕ does not admit a β -regular forward complete Finsler metric with an odd number of β -class geodesic strings.

The following questions seem to be open.

Question 3. Is the finite type assumption necessary for Corollary 2.27 in the case of surfaces? The answer is likely no, as the intuition of Nielsen-Thurston classification (it cannot be directly applied in the infinite type setting) suggests that ϕ should have pseudo-Anosov type behavior which an isometry even on an infinite type surface should not have.

3. COMPACTNESS PRELIMINARIES

Lemma 3.1. *Suppose that g is a complete metric on X , $\beta \in \pi_1^{\text{inc}}(X)$ and let $S \subset L_\beta X$ be a subset on which the g -length functional is bounded from above. Then the images in X of elements of S are contained in a fixed compact subset of X .*

Proof. Suppose otherwise. Fix an exhaustion by nested compact sets

$$\bigcup_{i \in \mathbb{N}} K_i = X, \quad K_i^\circ \supset K_{i-1}.$$

Then either there is sequence $\{o_i\}_{i \in \mathbb{N}}$, $o_i \in S$ such that $o_i \in K_i^\circ$, for K_i° the complement of K_i , which contradicts the fact that β is end incompressible. Or there is a sequence $\{o_k\}_{k \in \mathbb{N}}$, $o_k \in S$ such that:

- (1) Each o_k intersects K_{i_0} for some i_0 fixed.
- (2) For each $i \in \mathbb{N}$ there is a $k_i > i$ such that o_{k_i} is not contained in K_i .

Now if $\text{diam}(o_k)$ is bounded in k , then condition 1 implies that o_k are contained in a set of bounded diameter. (Here $\text{diam}(o_k)$ denotes the diameter of image o_k .) Consequently, by Hopf-Rinow theorem [8], o_k are contained in a compact set. But this contradicts condition 2, and the fact that K_i form an exhaustion of X .

Thus, we conclude that $\text{diam}(o_k)$ is unbounded, but this contradicts the hypothesis. $\square$

Lemma 3.2. *Let g be a complete metric on X , and let $\beta \in \pi_1^{\text{inc}}(X)$. Denote by $S_A(g, \beta) \subset S(g, \beta)$ the subset of elements with g -length at most A . Then $S_A(g, \beta) \subset S(g, \beta)$ is compact.*

Proof. By Lemma 3.1, the images of elements of $S_A(g, \beta) \subset S(g, \beta)$ are contained in compact K . Since elements of $S_A(g, \beta) \subset S(g, \beta)$ are parametrized by constant speed, and since we have a length bound $S_A(g, \beta)$ is an equicontinuous collection. The result then follows by the Arzelà–Ascoli theorem and standard arguments. $\square$

Corollary 3.3. *Suppose that g is a complete metric on X , all of whose class $\beta \in \pi_1^{\text{inc}}(X)$ geodesics are minimizing, then g is β -taut.*

Lemma 3.4. *Let g be a complete metric on X , and let $\beta \in \pi_1^{\text{inc}}(X)$. Then there exists a closed g -geodesic minimizing length among all loops in the free homotopy class β .*

Proof. Suppose otherwise and let

$$A = \inf_{\gamma \in L_{\beta} X} \ell_g(\gamma).$$

By the hypothesis that there is no geodesic with length A , and by Lemma 3.2, there is an $\epsilon > 0$ such that there is no closed geodesic o with $A \leq \ell_g(o) < A + \epsilon$.

By Lemma 3.1 the subset $M_{\beta, A} \subset L_{\beta} X$ consisting of elements with length less than $A + \epsilon$ is contained in a compact K . Pick $o \in M_{\beta, A}$ and apply the curve shortening process to it. Let $o_k, k \in \mathbb{N}$ be the stages of this process. Then image $o_k \subset K$. In particular $\{o_k\}$ must converge to a geodesic, for instance [36] (we can also argue via Arzelà–Ascoli to prove convergence) but this contradicts the condition on ϵ . $\square$

3.1. Proof of Theorem 2.5. Let $\{g_t\}, t \in [0, 1]$ be as in the hypothesis, with

$$(3.5) \quad \sup_t \left| \sup_{o \in S(g_t, \beta)} \ell_{g_t}(o) - \inf_{o \in S(g_t, \beta)} \ell_{g_t}(o) \right| < C,$$

and suppose that

$$\sup_{(o, t) \in S(\{g_t\}, \beta)} \ell_{g_t}(o) = \infty.$$

Then we have a sequence $\{o_k\}, k \in \mathbb{N}$, of closed class β g_{t_k} -geodesics in X , satisfying:

- (1) $\lim_{k \rightarrow \infty} t_k = t_{\infty} \in [0, 1]$.
- (2) $\lim_{k \rightarrow \infty} \ell_{g_{t_k}}(o_k) = \infty$, where $\ell_{g_{t_k}}(o_k)$ is the length with respect to g_{t_k} .

Applying Lemma 3.4, we may choose a length-minimizing closed $g_{t_{\infty}}$ -geodesic o_{∞} in the class β . And let L denote its $g_{t_{\infty}}$ -length. Let g_{aux} be a fixed auxiliary metric on X , and let L_{aux} be the g_{aux} length of o_{∞} .

Define a pseudo-metric d_{C^0} on the space of metrics on X as follows. Set $K = \text{image } o_{\infty}$. And set

$$V = \{v \in TX \mid \pi(v) \in K \text{ for } \pi : TX \rightarrow X \text{ the canonical projection, and } |v|_{aux} = 1\},$$

where $|v|_{aux}$ is the norm taken with respect to g_{aux} .

Then define:

$$d_{C^0}(g_1, g_2) = \sup_{v \in V} ||v|_{g_1} - |v|_{g_2}|.$$

By Properties 1 and 2, for every $\epsilon > 0$ we may find a $k > 0$ such that:

$$(3.6) \quad d_{C^0}(g_{t_k}, g_{t_{\infty}}) < \epsilon$$

and

$$(3.7) \quad \ell_{g_{t_k}}(o_k) > C + L + L_{aux} \cdot \epsilon.$$

By (3.6), we have:

$$\ell_{g_{t_k}}(o_{\infty}) < \ell_{g_{t_{\infty}}}(o_{\infty}) + L_{aux} \cdot \epsilon = L + L_{aux} \cdot \epsilon.$$

Combining with (3.7) we get:

$$\ell_{g_{t_k}}(o_k) > \ell_{g_{t_k}}(o_\infty) + C.$$

Since we may find a closed g_{t_k} -geodesic o' satisfying $\ell_{g_{t_k}}(o') \leq \ell_{g_{t_k}}(o_\infty)$, we get that

$$\left| \max_{o \in S(g_{t_k}, \beta)} \ell_{g_{t_k}}(o) - \min_{o \in S(g_{t_k}, \beta)} \ell_{g_{t_k}}(o) \right| > C,$$

and so we are in contradiction.

Thus,

$$\sup_{(o,t) \in \mathcal{O}(\{g_t\}, \beta)} \ell_{g_t}(o) < \infty.$$

Then compactness of $\mathcal{S}(\{g_t\}, \beta)$ follows by a slight modification of the proof of Lemma 3.2. $\square$

4. PRELIMINARIES ON REEB FLOW

Let (C^{2n+1}, λ) be a contact manifold with λ a contact form, that is a one form such that $\lambda \wedge (d\lambda)^n \neq 0$. Denote by R^λ the Reeb vector field satisfying:

$$d\lambda(R^\lambda, \cdot) = 0, \quad \lambda(R^\lambda) = 1.$$

Recall that a **closed λ -Reeb orbit** (or just Reeb orbit when λ is implicit) is a smooth map

$$o : (S^1 = \mathbb{R}/\mathbb{Z}) \rightarrow C$$

such that

$$\dot{o}(t) = cR^\lambda(o(t)),$$

with $\dot{o}(t)$ denoting the time derivative, for some $c > 0$ called the period. Let $S(R^\lambda, \beta)$ denote the space of all closed λ -Reeb orbits in free homotopy class β , with its compact open topology, and set

$$\mathcal{O}(R^\lambda, \beta) = S(R^\lambda, \beta)/S^1,$$

where $S^1 = \mathbb{R}/\mathbb{Z}$ acts by reparametrization $t \cdot o(\tau) = o(t + \tau)$.

5. DEFINITION OF THE FUNCTIONAL F AND PROOFS OF AUXILIARY RESULTS

Let X be a manifold. Fix once and for all an auxiliary Riemannian metric g_0 , and set

$$C = S_{g_0}^* X.$$

For any Riemann-Finsler metric g , let F_g^* denote the dual norm, and let

$$\rho_g : C \rightarrow S_g^* X, \quad \rho_g(\xi) = \frac{\xi}{F_g^*(\xi)}$$

be the fiberwise radial projection. We define a contact form on the fixed manifold C by

$$\lambda_g = \rho_g^* \lambda_{\text{can}},$$

where λ_{can} is the Liouville one-form on T^*X . We write

$$R^{\lambda_g}$$

for the Reeb vector field of λ_g on this fixed manifold C . Under ρ_g , the flow of R^{λ_g} corresponds to the g -geodesic flow.

The same convention applies to the Sasaki metric. If g is Riemannian, let g_S^{nat} denote the usual Sasaki metric on $S_g^* X$, and set

$$g_S = \rho_g^* g_S^{\text{nat}}$$

on the fixed manifold C . Similarly, all metric-dependent functions on $S_g^* X$, such as verticality functions, are regarded as functions on C by pullback through ρ_g . This fixed- C convention will be used throughout.

5.1. Lifted geodesic strings. We recall how closed geodesics in X are lifted to closed Reeb orbits in its cosphere bundle. Let C be the fixed cosphere model used in this paper, and let

$$q : C \rightarrow X$$

denote the projection. For a Riemann-Finsler metric g , the contact form λ_g on C has Reeb vector field R^{λ_g} . Its closed orbits project to oriented closed constant-speed g -geodesics in X .

Conversely, every oriented closed constant-speed g -geodesic has a unique Reeb lift to C . In the Riemannian case, if

$$o : S^1 \rightarrow X$$

is a nonconstant closed g -geodesic, its lift to the g -unit cotangent bundle is

$$\tilde{o}_g(t) = \frac{g_{o(t)}(\dot{o}(t), \cdot)}{|\dot{o}(t)|_g} \in S_g^*X.$$

Since o is a geodesic, this covector field is parallel along o . Thus, under the Levi-Civita splitting of $T(T^*X)$, the curve $\tilde{o}_g$ is horizontal. Hence, after the constant reparametrization determined by the speed of o , it is a trajectory of the Reeb vector field. Passing to the fixed model C by radial projection gives the corresponding closed R^{λ_g} -orbit in C . The same statement holds in the Riemann-Finsler case using the Legendre transform.

The projection q induces a homeomorphism between closed R^{λ_g} -orbit strings in C and oriented closed constant-speed g -geodesic strings in X . Hence, for every class $\beta \in \pi_1^{\text{inc}}(X)$, if $\mathcal{O}(g, \beta)$ denotes the class β g -geodesic string set as usual, then its Reeb lift

$$\tilde{\mathcal{O}}(g, \beta) = \{\tilde{o} \mid o \in \mathcal{O}(g, \beta)\} \subset \mathcal{O}(R^{\lambda_g})$$

is homeomorphic to $\mathcal{O}(g, \beta)$. In particular, if $\mathcal{O}(g, \beta)$ is compact, then $\tilde{\mathcal{O}}(g, \beta)$ is compact. Moreover, $\tilde{\mathcal{O}}(g, \beta)$ is open in $\mathcal{O}(R^{\lambda_g})$, since the projected free homotopy class in X is locally constant on the loop space of C .

5.2. Definition via the Fuller index. Suppose then that g is β -taut. Define

$$F(g, \beta) = i(\tilde{\mathcal{O}}(g, \beta), R^{\lambda_g}) \in \mathbb{Q},$$

where the right hand side is the Fuller index, see Appendix A.

For a path of metrics $\{g_t\}_{t \in [0,1]}$, the lifted cobordism of orbit string spaces

$$\tilde{\mathcal{O}}(\{g_t\}, \beta) = \{(\tilde{o}, t) \mid o \in \mathcal{O}(g_t, \beta)\}$$

is likewise homeomorphic to the downstairs cobordism $\mathcal{O}(\{g_t\}, \beta)$.

Proof of Theorem 2.7. Let $\beta \in \pi_1^{\text{inc}}(X)$ be given, and let $\{g_t\}_{t \in [0,1]}$ be a β -taut deformation. As discussed above, the lifted cobordism $\tilde{\mathcal{O}}(\{g_t\}, \beta)$ is compact by β -tautness. It is also open in the orbit cobordism, since the projected free homotopy class in X is locally constant. Basic invariance of the Fuller index, namely (A.4), then immediately gives

$$F(g_0, \beta) = F(g_1, \beta).$$

□

6. PROOF OF THEOREM 2.16

For simplicity, as the full argument is already lengthy, we prove the theorem under the assumption that g is Riemannian. In the Finsler case we must use a more modern variational setup for the Hilbert manifold of H^1 -loops. In the Finsler case the energy functional is generally only C^1 , rather than C^2 , on the Hilbert loop space. Nevertheless the required Palais–Smale compactness and critical point theory for closed Finsler geodesics are standard; see Caponio–Javaloyes–Masiello [18], Mercuri [34], and Lu [32]. With these replacements, the compactness argument for end-incompressible classes, and the Fuller-index invariance argument go through without substantial change.

The circle S^1 acts on $L_\beta X$ by reparametrization:

$$(s \cdot \gamma)(t) = \gamma(t + s).$$

If some $\gamma \in L_\beta X$ had nontrivial stabilizer, then γ would factor through a nontrivial covering $S^1 \rightarrow S^1$, contradicting indivisibility. Therefore the S^1 -action on $L_\beta X$ is free. Hence

$$H_*^{S^1}(L_\beta X, \mathbb{Z}) \cong H_*(L_\beta X/S^1, \mathbb{Z}),$$

and we will construct a finite total rank model for the latter.

Let

$$\mathcal{L}_\beta X = W^{1,2}(S^1, X)_\beta$$

be the Hilbert manifold of H^1 -loops in the free homotopy class β . Consider the energy functional

$$\mathcal{E}_g : \mathcal{L}_\beta X \rightarrow \mathbb{R}, \quad \mathcal{E}_g(\gamma) = \int_{S^1} |\dot{\gamma}(t)|_g^2 dt.$$

Since $S(g, \beta)$ is compact, $\mathcal{E}_g$ is bounded above on the critical set of $\mathcal{E}_g$ in class β . Choose

$$E_0 > \max_{\gamma \in S(g, \beta)} \mathcal{E}_g(\gamma) = M^2,$$

which exists by the β -tautness assumption. Set

$$\mathcal{U}_{E_0} = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_g(\gamma) < E_0\}.$$

Lemma 6.1. *The inclusion*

$$\mathcal{U}_{E_0} \hookrightarrow \mathcal{L}_\beta X$$

is an S^1 -equivariant homotopy equivalence. Consequently,

$$\mathcal{U}_{E_0}/S^1 \simeq \mathcal{L}_\beta X/S^1.$$

Moreover, the inclusion of smooth loops into the Hilbert loop space is an S^1 -equivariant homotopy equivalence, and hence

$$\mathcal{L}_\beta X/S^1 \simeq L_\beta X/S^1.$$

Therefore

$$\mathcal{U}_{E_0}/S^1 \simeq L_\beta X/S^1.$$

Proof. The energy functional $\mathcal{E}_g$ is smooth and S^1 -invariant. We equip $\mathcal{L}_\beta X$ with the standard S^1 -invariant Hilbert metric induced by g . With respect to this metric, the negative gradient flow of $\mathcal{E}_g$ is S^1 -equivariant.

We first recall the Palais–Smale compactness needed for the deformation lemma. Let $\{\gamma_j\} \subset \mathcal{L}_\beta X$ be a Palais–Smale sequence for $\mathcal{E}_g$ with

$$\mathcal{E}_g(\gamma_j) \leq A.$$

Then

$$\ell_g(\gamma_j) \leq \mathcal{E}_g(\gamma_j)^{1/2} \leq A^{1/2}.$$

Since $\beta \in \pi_1^{\text{inc}}(X)$, Lemma 3.1 implies that the images $\gamma_j(S^1)$ lie in a fixed compact subset of X . On this compact set, the usual Hilbert-loop compactness argument for the geodesic energy gives a strongly convergent subsequence in $\mathcal{L}_\beta X$. Thus $\mathcal{E}_g$ satisfies the Palais–Smale condition on sublevels.

By the choice of E_0 , the functional $\mathcal{E}_g$ has no critical points in

$$\{\mathcal{E}_g \geq E_0\}.$$

Therefore, for every $A > E_0$, Palais’ deformation lemma gives an S^1 -equivariant homotopy equivalence

$$\{\mathcal{E}_g < E_0\} \hookrightarrow \{\mathcal{E}_g < A\}.$$

Letting $A \rightarrow \infty$, or equivalently applying the negative gradient flow on the exhaustion by sublevels, we obtain an S^1 -equivariant homotopy equivalence

$$\mathcal{U}_{E_0} \simeq_{S^1} \mathcal{L}_\beta X.$$

Passing to quotients gives

$$\mathcal{U}_{E_0}/S^1 \simeq \mathcal{L}_\beta X/S^1.$$

Finally, the inclusion

$$C^\infty(S^1, X)_\beta \hookrightarrow W^{1,2}(S^1, X)_\beta$$

is an S^1 -equivariant homotopy equivalence by standard equivariant smoothing approximation for loops. Since $L_\beta X$ denotes the same free homotopy component in the smooth or compact-open loop space, we get

$$\mathcal{L}_\beta X \simeq_{S^1} L_\beta X.$$

Combining the quotient equivalences gives

$$\mathcal{U}_{E_0}/S^1 \simeq L_\beta X/S^1.$$

□

We now compare the Fuller index with the equivariant Morse model. The only subtlety is that after perturbing g to a regular metric g' , the full class β geodesic set of g' need not be compact. We therefore work in an isolating neighborhood of the compact lifted orbit block coming from g .

Let g_S denote the Sasaki metric on the fixed contact manifold C induced by g under the convention of Section 5. Define the S^1 -invariant loop set $\widehat{U}_{\sqrt{E_0}}$ to consist of all loops $\gamma : S^1 \rightarrow C$ such that

$$[q \circ \gamma] = \beta, \quad \ell_{g_S}(\gamma) < \sqrt{E_0}.$$

Let

$$U_{\sqrt{E_0}} = \widehat{U}_{\sqrt{E_0}}/S^1$$

be its image in the orbit-string quotient. By (A.8) and the choice of E_0 , this is an isolating neighborhood of

$$N_0 = \widetilde{\mathcal{O}}(g, \beta)$$

in the sense of Lemma A.5. It is also g_S -controlled, in the sense of Definition A.1, with the lower length bound supplied by the nontrivial projected class β and K provided by Lemma 3.1.

Let W be as in Lemma A.5, and let ϵ be the corresponding Fuller constant for the isolating data

$$(N_0, U_{\sqrt{E_0}}, W).$$

Choose a β -regular complete metric g' sufficiently C^2 -close ϵ' -close to g so that, on the fixed manifold C , the vector field $R^{\lambda_{g'}}$ is C^0 -close to R^{λ_g} . Choose E' with

$$\max_{\gamma \in N_0} \ell_{g_S}(\gamma) < \sqrt{E'} < \sqrt{E_0}.$$

Set $\widehat{U}$ to be the set of loops $\gamma : S^1 \rightarrow C$ such that

$$[q \circ \gamma] = \beta, \quad \ell_{g'_S}(\gamma) < \sqrt{E'},$$

and set

$$U = \widehat{U}/S^1.$$

After shrinking the perturbation if necessary, we may assume

$$W \subset U \subset U_{\sqrt{E_0}}.$$

Thus Lemma A.7 gives

$$F(g, \beta) = i(N_1, R^{\lambda_{g'}}),$$

where

$$N_1 = U \cap \mathcal{O}(R^{\lambda_{g'}}).$$

It remains to compute the last Fuller index. Let

$$\mathcal{U}_{E'}^{g'} = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_{g'}(\gamma) < E'\}.$$

For a constant-speed g' -geodesic, its canonical lift has g'_S -length equal to its g' -length. Hence the critical circles of $\mathcal{E}_{g'}$ in $\mathcal{U}_{E'}^{g'}$ are in one-to-one correspondence with the elements of N_1 . Since g' is β -regular, this critical set is a finite union of nondegenerate critical circles

$$C_o \simeq S^1.$$

Although the S^1 -action on $\mathcal{L}_\beta X$ is only continuous, each individual rotation is a smooth isometry preserving $\mathcal{E}_{g'}$, so the negative gradient flow of $\mathcal{E}_{g'}$ commutes with the action, and the deformations below are S^1 -equivariant. The Morse–Bott handlebody $\mathcal{MB}$ obtained from the unstable disk bundles of these critical circles is S^1 -invariant, and the standard Hilbert-manifold Morse–Bott deformation theorem gives an S^1 -equivariant homotopy equivalence

$$\mathcal{MB} \simeq_{S^1} \mathcal{U}_{E'}^{g'}.$$

After passing to the quotient by S^1 , the unstable disk bundle over C_o contributes one cell of dimension $\text{morse}(o)$; here freeness of the action (indivisibility of β) is used, as the quotient of the disk bundle by the free S^1 -action is a single cell. Therefore

$$\chi(\mathcal{U}_{E'}^{g'}/S^1) = \chi(\mathcal{MB}/S^1) = \sum_{o \in N_1} (-1)^{\text{morse}(o)}.$$

We now compare the g' -sublevel with the Hilbert sublevel for g . Choose numbers a, b such that

$$M^2 < a < E' < b < E_0$$

and such that $\mathcal{E}_g$ has no critical points on

$$\{a \leq \mathcal{E}_g \leq b\}.$$

For ϵ' small enough we have $\mathcal{U}_a^g \subset \mathcal{U}_{E'}^{g'} \subset \mathcal{U}_b^g$, so that by Lemma 6.2 the inclusion

$$\mathcal{U}_{E'}^{g'} \hookrightarrow \mathcal{U}_b^g$$

is an S^1 -equivariant homotopy equivalence. Since all critical points of $\mathcal{E}_g$ have energy at most $M^2 < b$, the inclusion

$$\mathcal{U}_b^g \hookrightarrow \mathcal{U}_{E_0}^g$$

is also an S^1 -equivariant homotopy equivalence. Thus

$$\mathcal{U}_{E'}^{g'} \simeq_{S^1} \mathcal{U}_{E_0}^g.$$

By Lemma 6.1,

$$\mathcal{U}_{E_0}^g/S^1 \simeq L_\beta X/S^1.$$

Consequently

$$\chi(\mathcal{U}_{E'}^{g'}/S^1) = \chi(L_\beta X/S^1).$$

In particular $H_*^{S^1}(L_\beta X, \mathbb{Z})$ has finite total rank.

On the other hand, since g' is β -regular, the Fuller index formula, summed over the finitely many lifted free homotopy classes met by N_1 , gives

$$i(N_1, R^{\lambda_{g'}}) = \sum_{o \in N_1} \frac{(-1)^{\text{morse}(o)}}{\text{mult}(o)}.$$

Since β is indivisible, all multiplicities in this class are 1. Hence

$$i(N_1, R^{\lambda_{g'}}) = \chi(\mathcal{U}_{E'}^{g'}/S^1) = \chi(L_\beta X/S^1).$$

Therefore

$$F(g, \beta) = \chi(L_\beta X/S^1) = \chi^{S^1}(L_\beta X).$$

□

We state the following elementary lemma separately, since it may be of independent interest.

Lemma 6.2. *For a metric h and a number r , set*

$$\mathcal{U}_r^h = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_h(\gamma) < r\}.$$

Let $a < b$, and suppose that $\mathcal{E}_g$ satisfies the Palais–Smale condition on the band

$$\mathcal{B} = \{a \leq \mathcal{E}_g \leq b\}$$

and has no critical points there. Let g' be sufficiently C^2 -close to g , so that $\nabla \mathcal{E}_{g'}$ is C^0 -close to $\nabla \mathcal{E}_g$ on $\mathcal{B}$. Suppose

$$\mathcal{U}_a^g \subset \mathcal{C} \subset \mathcal{U}_b^g,$$

where

$$\mathcal{C} = \{\mathcal{E}_{g'} < c\}$$

is an S^1 -invariant sublevel set of $\mathcal{E}_{g'}$. Then the inclusion

$$\mathcal{C} \hookrightarrow \mathcal{U}_b^g$$

is an S^1 -equivariant homotopy equivalence.

Proof. By the Palais–Smale condition and the absence of critical points on $\mathcal{B}$, there exists $\mu > 0$ such that

$$(6.3) \quad \|\nabla \mathcal{E}_g\| \geq \mu$$

on $\mathcal{B}$. For g' sufficiently close to g , we have

$$(6.4) \quad \|\nabla \mathcal{E}_{g'} - \nabla \mathcal{E}_g\| < \mu/2$$

on the same band. Therefore

$$\langle \nabla \mathcal{E}_g, \nabla \mathcal{E}_{g'} \rangle > 0$$

on $\mathcal{B}$.

Since

$$\mathcal{U}_a^g \subset \mathcal{C} \subset \mathcal{U}_b^g,$$

we have

$$\mathcal{U}_b^g - \mathcal{C} \subset \mathcal{U}_b^g - \mathcal{U}_a^g \subset \mathcal{B}.$$

Now along a negative $\mathcal{E}_{g'}$ -gradient trajectory lying in $\mathcal{B}$ we have:

$$\frac{d}{dt} \mathcal{E}_g = -\langle \nabla \mathcal{E}_g, \nabla \mathcal{E}_{g'} \rangle < 0.$$

Note that the topological boundary of $\mathcal{U}_b^g - \mathcal{C}$ is contained in $\{\mathcal{E}_g = b\} \cup \{\mathcal{E}_{g'} = c\}$. Consequently the negative $\mathcal{E}_{g'}$ -gradient flow line of an element in $\mathcal{U}_b^g - \mathcal{C}$ cannot exit $\mathcal{U}_b^g$ before hitting $\mathcal{C}$. Moreover, $\mathcal{C}$ is forward-invariant because it is a sublevel set of $\mathcal{E}_{g'}$.

The estimates above also imply that $\mathcal{E}_{g'}$ has no critical points on $\mathcal{U}_b^g - \mathcal{C}$. The standard deformation argument of Palais [39, Section 10, Proposition (2)], applied to $\mathcal{E}_{g'}$ on the region $\mathcal{U}_b^g - \mathcal{C}$, therefore gives a deformation retraction of $\mathcal{U}_b^g$ onto $\mathcal{C}$. The energies and the Hilbert metrics are S^1 -invariant, so the gradient flow is S^1 -equivariant. Hence the deformation retraction is S^1 -equivariant. $\square$

7. PRODUCT FORMULA

In this section we prove a kind of product formula for the F -count. This is used in all our applications.

Theorem 7.1 (Product formula). *Let*

$$Z \hookrightarrow (X, g) \xrightarrow{p} (Y, g_Y)$$

be a β -periodic fibration, with Y closed, connected and such that all smooth closed contractible g_Y -geodesics in Y are constant. Then p is β -taut. Fix $y_0 \in Y$, set $Z_0 = p^{-1}(y_0)$, and choose

$$\beta_Z \in \pi_1^{\text{inc}}(Z_0)$$

such that

$$(i_{y_0})_*(\beta_Z) = \beta,$$

where $i_{y_0} : Z_0 \hookrightarrow X$ is the inclusion. Denote by g_{y_0} the restriction of g to the fiber of y_0 . Let

$$H = \pi_1(Y, y_0)$$

act on $\pi_1^{\text{inc}}(Z_{y_0})$ by holonomy, and set

$$S_{y_0} = H \cdot \beta_Z.$$

Then

$$F(g, \beta) = \text{card}(S_{y_0}) \cdot \chi(Y) \cdot F(g_{y_0}, \beta_Z).$$

Proof. We prove β -tautness. We first show that every class β g -geodesic in X is vertical. Let

$$o : S^1 \rightarrow X$$

be a closed g -geodesic representing the class β . Since β is a fiber class, the loop

$$p \circ o : S^1 \rightarrow Y$$

is contractible. Since p is a parallel Riemannian submersion, the local de Rham product decomposition for p implies that $p \circ o$ is a g_Y -geodesic, possibly constant. Hence $p \circ o$ is a smooth closed contractible g_Y -geodesic. By hypothesis, every such geodesic is constant. Therefore $p \circ o$ is constant, and so o is contained in a single fiber

$$Z_y = p^{-1}(y).$$

Thus the class β geodesic strings in X are precisely the vertical geodesic strings whose fiber class represents β in X .

It remains to prove compactness. Let o_j be a sequence of class β g -geodesic strings in X . By the previous paragraph, each o_j is contained in a fiber Z_{y_j} . Since Y is compact, after passing to a subsequence we may assume

$$y_j \rightarrow y_\infty.$$

Choose a contractible coordinate ball

$$U \subset Y$$

around y_∞ . For all large j , we have $y_j \in U$. Since p is a parallel Riemannian submersion, the local de Rham product theorem gives a local Riemannian product decomposition

$$p^{-1}(U) \cong Z_{y_\infty} \times U.$$

Under this identification, vertical geodesics in the fibers Z_{y_j} are identified with geodesics of the fiber metric on Z_{y_∞} , and their fiber free homotopy classes lie in the finite holonomy orbit

$$S_{y_\infty}.$$

By the fiber-tautness condition in the definition of a β -periodic fibration, the union of the geodesic string sets

$$\bigcup_{\alpha \in S_{y_\infty}} S(g_{y_\infty}, \alpha)$$

is compact. Since S_{y_∞} is finite, this is a finite union of compact sets. Therefore, after passing to a further subsequence, the vertical geodesic strings o_j , transported to the model fiber Z_{y_∞} , converge to a fiber geodesic string. Transporting back through the local product chart gives convergence of the original o_j in the class β geodesic string space of X .

Thus every sequence of class β geodesic strings has a convergent subsequence. Hence the class β geodesic string set is compact, and g is β -taut.

Let

$$C = S^*X$$

be the unit cotangent bundle of (X, g) , with projection

$$\text{pr} : C \rightarrow X.$$

For this fixed metric calculation, this is equivalent to the fixed- C convention of Section 5. Let R^{λ_g} denote the Reeb vector field on C . The F -invariant is computed from the compact lifted orbit set

$$N_0 = \tilde{\mathcal{O}}(g, \beta).$$

The idea is to localize contributions over critical points of a Morse function on Y .

Let

$$\pi^{\text{vert}} : TX \rightarrow T^{\text{vert}}X = \ker(dp)$$

be the g -orthogonal projection. By assumptions on $p : X \rightarrow Y$,

$$\nabla \pi^{\text{vert}} = 0,$$

where π^{vert} is viewed as a $(1, 1)$ -tensor field.

For $\xi \in C$, let

$$v_\xi \in T_{\text{pr}(\xi)}X$$

be the g -dual unit vector to ξ . Define

$$P : C \rightarrow \mathbb{R}$$

by

$$P(\xi) = |\pi^{\text{vert}} v_\xi|_g^2.$$

Thus $P = 1$ on vertical unit covectors and $P = 0$ on horizontal unit covectors.

Let $f : Y \rightarrow \mathbb{R}$ be a Morse function. Define

$$\tilde{f} : C \rightarrow \mathbb{R}$$

by

$$\tilde{f} = f \circ p \circ \text{pr}.$$

Equip $C = S^*X$ with the Sasaki metric g_S induced by g , as in the Proof of Theorem 2.16. We shall use the following two elementary identities.

First,

$$(7.2) \quad dP(R^{\lambda_g}) = 0.$$

Indeed, let $\xi(\tau)$ be a flow line of R^{λ_g} , and set

$$x(\tau) = \text{pr}(\xi(\tau)).$$

Then $x(\tau)$ is a unit-speed g -geodesic. Let

$$v(\tau) = \dot{x}(\tau) = v_{\xi(\tau)}$$

be its velocity vector field. Since $x(\tau)$ is a geodesic,

$$\nabla_{\tau} v = 0.$$

Since $\nabla \pi^{\text{vert}} = 0$, we have

$$\nabla_{\tau}(\pi^{\text{vert}} v) = \pi^{\text{vert}}(\nabla_{\tau} v) = 0.$$

Therefore

$$\begin{aligned} \frac{d}{d\tau} P(\xi(\tau)) &= \frac{d}{d\tau} |\pi^{\text{vert}} v(\tau)|_g^2 \\ &= 2 \langle \nabla_{\tau}(\pi^{\text{vert}} v), \pi^{\text{vert}} v \rangle_g \\ &= 0. \end{aligned}$$

Hence $dP(R^{\lambda_g}) = 0$.

Second,

$$(7.3) \quad dP(\nabla_{g_S} \tilde{f}) = 0.$$

Since $\tilde{f} = f \circ p \circ \text{pr}$ depends only on the base point $\text{pr}(\xi)$:

$$\nabla_{g_S} \tilde{f} \in T^{\text{hor}} C.$$

More precisely, $\nabla_{g_S} \tilde{f}$ is the horizontal lift of

$$\nabla_g(f \circ p)$$

on X . Let $\xi(s)$ be a horizontal curve in C , and set

$$x(s) = \text{pr}(\xi(s)).$$

By definition of the Levi-Civita horizontal distribution on S^*X , the covectors $\xi(s)$ are parallel along $x(s)$. Equivalently, their g -dual vectors

$$v(s) = v_{\xi(s)}$$

are parallel along $x(s)$:

$$\nabla_s v(s) = 0.$$

Using again $\nabla \pi^{\text{vert}} = 0$, we get

$$\nabla_s(\pi^{\text{vert}} v(s)) = \pi^{\text{vert}}(\nabla_s v(s)) = 0.$$

Hence

$$\begin{aligned} \frac{d}{ds} P(\xi(s)) &= \frac{d}{ds} |\pi^{\text{vert}} v(s)|_g^2 \\ &= 2 \langle \nabla_s(\pi^{\text{vert}} v(s)), \pi^{\text{vert}} v(s) \rangle_g \\ &= 0. \end{aligned}$$

Taking $\dot{\xi}(0) = \nabla_{g_S} \tilde{f}(\xi)$ gives

$$dP_{\xi}(\nabla_{g_S} \tilde{f}) = 0.$$

Since ξ was arbitrary,

$$dP(\nabla_{g_S} \tilde{f}) = 0.$$

Now set

$$V_{\varepsilon} = R^{\lambda_g} - \varepsilon \nabla_{g_S}(P + \tilde{f}).$$

Recall first the discussion of Fuller continuation in the Appendix. The set $N_0 = \tilde{\mathcal{O}}(g, \beta)$ is compact by β -tautness. Choose a controlled isolating neighborhood U of N_0 and an open neighborhood W with

$\overline{W} \subset U$, and let ϵ' be the corresponding Fuller constant. Take $\epsilon > 0$ small enough so that V_ϵ is C^0 ϵ' -close to R^{λ_g} on the compact region determined by U . Lemma A.7 gives a compact continued set N_ϵ and

$$F(g, \beta) = i(N_\epsilon, V_\epsilon).$$

Note that the above does not require any further properties of V_ϵ , such as completeness. We set up the Fuller index theory to sidestep such issues.

We now identify the closed orbits in N_ϵ . We show that P is a Lyapunov function for V_ϵ . Mainly, along a trajectory $\xi(\tau)$ of V_ϵ , we have

$$\begin{aligned} \frac{d}{d\tau} P(\xi(\tau)) &= dP(R^{\lambda_g}) - \epsilon dP(\nabla_{g_S} P) - \epsilon dP(\nabla_{g_S} \tilde{f}) \\ &= -\epsilon |\nabla_{g_S} P|_{g_S}^2 \leq 0. \end{aligned}$$

Thus P is nonincreasing along the V_ϵ -flow.

If $\xi(\tau)$ is a closed orbit of V_ϵ , then $P(\xi(\tau))$ returns to its initial value after one period. Since P is nonincreasing, it must be constant along the orbit. Hence

$$\nabla_{g_S} P = 0$$

along the orbit.

On each unit cotangent sphere, P is the squared norm of the vertical component of the corresponding unit vector. Therefore the critical locus of P consists of the two critical loci

$$P = 1 \quad \text{and} \quad P = 0,$$

corresponding respectively to vertical and horizontal covectors. At $\epsilon = 0$, the lifted orbit set under consideration is vertical, hence lies in $P = 1$. Taking the controlled neighborhood U (above) so that all its elements are disjoint (as loops) from $P = 0$, we get that every closed orbit in N_ϵ lies in the vertical critical locus $P = 1$. Consequently, every closed orbit in N_ϵ projects to a geodesic contained in a fiber Z_y .

It remains to see over which fibers these vertical orbits occur. On $P = 1$, the Reeb vector field restricts to the geodesic Reeb field of the fiber (Z_y, g_y) , while $\tilde{f}$ depends only on the base point y . The perturbation

$$-\epsilon \nabla_{g_S} \tilde{f}$$

moves the base point in the negative gradient direction of f . Therefore a vertical closed orbit of V_ϵ can occur only over a critical point of f . Conversely, if $y \in \text{Crit}(f)$, then

$$\nabla_{g_S} \tilde{f} = 0$$

over Z_y , and the vertical part of V_ϵ is just the fiber geodesic flow, up to the harmless P -term which vanishes on the vertical critical locus. Thus the closed orbits of V_ϵ in N_ϵ are precisely the vertical closed geodesic orbits lying in fibers over critical points of f .

We now compute the index. For $y \in Y$, put

$$H_y = \pi_1(Y, y),$$

acting on $\pi_1^{\text{inc}}(Z_y)$ by holonomy. Choose

$$\beta_Z \in \pi_1^{\text{inc}}(Z_y)$$

with

$$(i_y)_*(\beta_Z) = \beta.$$

Set

$$S_y = H_y \cdot \beta_Z.$$

The Fuller index is additive over disjoint compact open subsets. Thus it remains to compute the local index contribution $i(N_{\varepsilon,y}, V_\varepsilon)$ corresponding to the isolated subset of N_ε lying over a critical point y of f . We defer this computation to the Appendix as Lemma B.1; it yields that

$$(7.4) \quad i(N_{\varepsilon,y}, V_\varepsilon) = (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha),$$

Next, we show that $F(g_y, \cdot)$ is constant on S_y . Indeed, let

$$\gamma : [0, 1] \rightarrow Y$$

be a path based at y . Pulling back $p : X \rightarrow Y$ along γ gives a fibration over $[0, 1]$. By the same proof that p is β -taut, this pullback is a taut deformation of the fiber metric from (Z_y, g_y) back to itself, with final fiber identified with the initial one by the holonomy map $h_\gamma : Z_y \rightarrow Z_y$. Fuller index invariance under taut deformation gives

$$F(g_y, \alpha) = F(g_y, h_{\gamma,*}\alpha).$$

Thus $F(g_y, \alpha)$ is constant on the holonomy orbit S_y . Hence

$$\sum_{\alpha \in S_y} F(g_y, \alpha) = \#S_y \cdot F(g_y, \beta_Z).$$

If $y, y' \in Y$, holonomy along any path from y to y' identifies S_y with $S_{y'}$. The same path gives a taut deformation between the fiber metrics, so

$$\sum_{\alpha \in S_y} F(g_y, \alpha)$$

is independent of y . So we get:

$$\begin{aligned} F(g, \beta) &= \sum_{y \in \text{Crit}(f)} i(N_{\varepsilon,y}, V_\varepsilon) \\ &= \sum_{y \in \text{Crit}(f)} (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha), \\ &= \sum_{y \in \text{Crit}(f)} (-1)^{\text{morse}(y)} \cdot \#S_{y_0} \cdot F(g_{y_0}, \beta_Z) \\ &= \chi(Y) \cdot \#S_{y_0} \cdot F(g_{y_0}, \beta_Z). \end{aligned}$$

□

Corollary 7.5. *Let $X = Z \times Y$, with Y closed, connected, such that all smooth closed contractible g_Y -geodesics in Y are constant and such that Z, g_Z is taut. Let $\beta \in \pi_1^{\text{inc}}(X)$ be a fiber class. Then $g = g_Z \times g_Y$ is β -taut. Furthermore,*

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z).$$

Proof. This is immediate from the theorem. □

8. PROOF OF THEOREM 2.23 AND ITS COROLLARIES

We prove the first part. By Cartan–Hadamard, a complete negatively curved Riemannian manifold has at most one closed geodesic in each nontrivial free homotopy class; moreover the absence of nontrivial periodic Jacobi fields in negative curvature gives nondegeneracy. Existence of a geodesic in the end-incompressible class β follows from Lemma 3.4. Hence a complete negatively curved metric would give a unique nondegenerate class β geodesic string. It thus suffices to prove the second part.

We prove the second part. Suppose first that β is atomic, so that β has a representative that is a k -cover of a loop, $k \geq 1$, whose corresponding class α is indivisible. By the assumption that all contractible g_Y

geodesics are constant, classical Morse theory Milnor [36] tells us that Y has vanishing higher homotopy groups $\pi_j(Y, y_0)$, $j \geq 2$. Thus by the long exact sequence of a fibration $i_{Z,*} : \pi_1(Z, z_0) \rightarrow \pi_1(X, i_Z(z_0))$ is a group injection, and image $i_{Z,*} = \ker p_*$. Now $\pi_1(Y, y_0)$ is torsion-free since it is a finite dimensional $K(\pi_1(Y, y_0), 1)$ space, Hatcher [24, Prop. 2.45]. It follows that α is also a fiber class since

$$p_*(\alpha^k) = 0 \implies p_*(\alpha) = 0.$$

Furthermore, $\alpha \in \pi_1^{inc}(X)$ (otherwise β isn't).

Lemma 8.1. *Let*

$$Z \hookrightarrow (X, g) \xrightarrow{p} (Y, g_Y)$$

be a g -parallel Riemannian submersion with complete total space, and assume that Y is connected with $\pi_2(Y) = 0$. Let

$$\beta, \gamma \in \pi_1^{inc}(X)$$

be fiber classes such that

$$\beta = \gamma^m$$

for some $m \geq 1$. If g is β -taut, then g is γ -taut and p is γ -holonomy-finite.

Proof. We show γ -tautness. Every closed γ -geodesic o_γ has m -fold cover $o_\gamma^m \in S(g, \beta)$, so

$$\ell_g(o_\gamma) = \frac{1}{m} \ell_g(o_\gamma^m) \leq A/m, \quad A := \max_{o \in S(g, \beta)} \ell_g(o) < \infty,$$

where finiteness of A is β -tautness. Hence $S(g, \gamma) = S_{A/m}(g, \gamma)$, which is compact by Lemma 3.2.

We show γ -holonomy-finiteness. Choose

$$\gamma_Z \in \mathcal{R}(\gamma)$$

and set

$$S_{y_0} = \pi_1(Y, y_0) \cdot \gamma_Z.$$

Suppose, toward a contradiction, that S_{y_0} is infinite. For each

$$\gamma_i \in S_{y_0}$$

choose a closed g_{y_0} -geodesic

$$o_i \subset Z_{y_0}$$

representing γ_i . Such geodesics exist by Lemma 3.4. Since the classes γ_i are distinct in Z_{y_0} , the loops o_i are pairwise distinct as elements of the fiber loop space.

We first record a separation property for these loops in the ambient loop space LX . Choose a contractible trivializing neighborhood

$$U \subset Y$$

of y_0 . For a loop $o \subset Z_{y_0}$, let

$$\mathcal{V}_{U,o} \subset LX$$

be the set of loops $\eta : S^1 \rightarrow X$ such that

$$p(\eta(S^1)) \subset U$$

and such that η , regarded as a loop in $p^{-1}(U)$, is freely homotopic in $p^{-1}(U)$ to o . The set $\mathcal{V}_{U,o}$ is open in LX : the condition that the image lie in $p^{-1}(U)$ is open, and free homotopy classes are open components of the loop space of $p^{-1}(U)$.

Moreover, if o_i and o_j represent distinct free homotopy classes in Z_{y_0} , then

$$\mathcal{V}_{U,o_i} \cap \mathcal{V}_{U,o_j} = \emptyset.$$

Indeed, an element of the intersection would give a free homotopy between o_i and o_j inside $p^{-1}(U)$. Since U is contractible, the inclusion

$$Z_{y_0} \hookrightarrow p^{-1}(U)$$

is a homotopy equivalence, and hence o_i and o_j would already be freely homotopic inside Z_{y_0} , contrary to construction.

On the other hand, $S(g, \gamma)$ is compact. Thus the infinite family $\{o_i\} \subset S(g, \gamma)$ has an accumulation point $o_\infty \in S(g, \gamma)$. Since every o_i lies in the fiber Z_{y_0} , the limit o_∞ also lies in Z_{y_0} . The open neighborhood $\mathcal{V}_{U, o_\infty}$ of o_∞ therefore contains o_i for infinitely many i , forcing these o_i 's to lie in the same fiber free homotopy class as o_∞ . This contradicts the fact that the classes γ_i are pairwise distinct. Hence S_{y_0} is finite and p is γ -holonomy-finite. □

By Theorem 7.1 p is β -taut. By the lemma above p is α -holonomy finite and hence is an α -periodic fibration. Now, if $\chi(Y) \neq \pm 1$ then by the product formula of Theorem 7.1 $F(g, \alpha) \neq \pm 1$, since $F(g_{y_0}, \alpha_Z)$ is an integer by Theorem 2.16.

Now, if X admits a complete metric h with a unique and nondegenerate class β h -geodesic string o_β , then it clearly also has a unique class α h -geodesic string o_α . Furthermore, o_α must be nondegenerate as otherwise o_β , which is a k -fold cover of o_α is degenerate. Then we have:

$$F(h, \alpha) = \pm 1.$$

But this is a contradiction, since $F(h, \alpha) = F(g, \alpha)$ by Theorem 2.16.

We now prove the general case. Suppose by contradiction that X admits a complete metric h with a unique and nondegenerate class β h -geodesic string o_β . By the preceding paragraph, the class β is not atomic. Now o_β covers a multiplicity one geodesic string o_α in some class $\alpha \in \pi_1^{inc}(X)$. Moreover, o_α is the unique geodesic string in its class, otherwise o_β would not be unique in its class. We prove that α is indivisible, which will be a contradiction to β not being atomic and will complete the proof.

Suppose otherwise, so that $\alpha_{x_0} = \rho^k$ for $k > 1$ and $\rho \in \pi_1(X, x_0)$ (with the notation of Definition 2.14). Let o_ρ be a class ρ , closed h -geodesic string (where by slight abuse ρ also denotes the class in $\pi_1^{inc}(X)$ corresponding to the based class ρ .) It is immediate that the k cover of o_ρ , o_ρ^k represents α and is a h -geodesic string. By the uniqueness, $o_\alpha = o_\rho^k$. But this contradicts multiplicity one of o_α . So α is indivisible. □

Proof of Corollary 2.24. Let g_Z, g_Y be complete Riemannian metrics with g_Z taut, and g_Y not admitting contractible closed geodesics. Take the product metric $g = g_Z \times g_Y$ on $X = Z \times Y$. For a class $\beta \in \pi_1^{inc}(X)$ in the image of the inclusion $\pi_1^{inc}(Z) \rightarrow \pi_1^{inc}(X)$, the natural projection $X \rightarrow Y$ is automatically a β -periodic fibration. Then the conclusion readily follows from Theorem 2.23. □

Proof of Corollary 2.27. The first two parts follow from Theorem 2.23, Lemma 2.26 and Example 6. The last part is deduced as follows. The product formula Theorem 7.1 gives us $F(g_\phi, \beta) = 0$. So for any β -regular g with finitely many class β g -geodesic strings o , since β is indivisible, by Corollary 1.2 we have:

$$\chi^{S^1}(L_\beta X_\phi) = F(g, \beta) = \sum_o (-1)^{\text{morse } o} = 0.$$

Then the conclusion readily follows. □

Proof of Theorem 2.17. By Theorem 7.1 0 is certainly a value of the invariant F . We first prove that every negative rational number is the value of the invariant. Let p, q be positive integers. Let Y be a closed surface of genus $(p+1) > 1$ with a hyperbolic metric g_Y , let Z be the genus 2 closed surface with a hyperbolic metric g_Z and let $\beta_Z \in \pi_1^{inc}(Z)$ be the class represented by a $2 \cdot q$ -fold covering of a simple closed loop representing a generator of the fundamental group of Z .

Let $X = Y \times Z$ with the product metric $g = g_Y \times g_Z$ and $p : X \rightarrow Y$ the canonical projection. Taking $\beta = (i_Z)_*(\beta_Z)$, by Corollary 7.5,

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z) = (-2p) \cdot \frac{1}{2q} = -\frac{p}{q}.$$

So we proved our first claim.

Let again p, q be positive integers. Let Y be closed surface of genus 2, with a hyperbolic metric g_Y . And let Z be a manifold satisfying $F(g_Z, \beta_Z) = -\frac{p}{2q}$ for some β_Z -taut metric g_Z on Z and for some class $\beta_Z \in \pi_1^{inc}(Z)$. This exists by the discussion above. Let $g = g_Y \times g_Z$ be the product metric on $Y \times Z$, and β as above. Analogously to the discussion above we get:

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z) = (-2) \cdot \frac{-p}{2q} = \frac{p}{q}.$$

□

9. NEAR-FLAT RIGIDITY VIA KAM CONFINEMENT

We show that every Riemann-Finsler metric on T^2 sufficiently close to a flat metric g_0 is taut, with every class β closed geodesic of length at most $(1 + \varepsilon)$ times the minimal g_0 -length in the class, and is taut homotopic to g_0 .

Proposition 9.1 (Near-flat rigidity). *Let g_0 be a flat Riemannian metric on T^2 . There is an absolute $k_0 \in \mathbb{N}$ such that for every $\varepsilon \in (0, 1]$ there is $\delta > 0$ with the following property. If g' is a Riemann-Finsler metric on T^2 with $\|g' - g_0\|_{C^{k_0}} < \delta$, then for every nontrivial free homotopy class β , every closed g' -geodesic o of class β satisfies*

$$(9.2) \quad \ell_{g'}(o) \leq (1 + \varepsilon) L_\beta,$$

where L_β denotes the minimal g_0 -length in class β . Consequently g' is taut, and any two metrics in the δ -ball are joined by a taut deformation; in particular every such g' is taut homotopic to g_0 .

Note the order of quantifiers: δ does not depend on β . For Finsler g' , $\|g' - g_0\|_{C^{k_0}}$ denotes the C^{k_0} -distance of F'^2 to F_0^2 on $\{\frac{1}{2} \leq F_0 \leq 2\} \subset TT^2$.

The classical theory of minimal geodesics on the torus (Morse [37], Hedlund [25], Bangert [7], Mather [33]) controls, for every metric, the *minimizers* in each class. Proposition 9.1 is orthogonal to it: the bound is on *all* class β closed geodesics, not just the minimizers, and no minimizer-theoretic argument reaches it. The mechanism is confinement by invariant tori of the geodesic flow which are graphs over T^2 and hence, in pairs, separate the unit tangent bundle. Intrinsically this graph property is the second Birkhoff theorem, established for Lagrangian invariant tori by Bialy–Polterovich and Polterovich [14, 40, 11, 10] and, without a dynamical assumption, by Arnaud [5]; for the merely continuous invariant tori of T^2 geodesic flows it is due to Bialy [12] (see also [13]). In the perturbative regime below it comes for free from KAM. E. Hopf's rigidity theorem [28]— ST^2 is foliated by incompressible invariant tori if and only if the metric is flat—explains why the smallness of δ cannot be removed: total confinement characterizes flatness, and away from g_0 one has only the Cantor family of surviving tori that KAM provides, which suffices for (9.2).

9.1. Setup and normalization. Every flat torus is isometric to $\mathbb{R}^2/\Lambda$ with the Euclidean metric; composing with a linear identification $\mathbb{R}^2/\mathbb{Z}^2 \cong \mathbb{R}^2/\Lambda$ produces a diffeomorphism ϕ of T^2 with ϕ^*g_0 constant-coefficient. Pulling the problem back by ϕ distorts C^{k_0} -balls by a fixed constant and preserves geodesics, lengths and classes. So assume $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ and $g_0 = G_0$, a constant positive-definite matrix; to simplify notation further we take G_0 to be the identity, so that g_0 is the standard flat metric. The general case is identical, with the constants below depending on G_0 ; see also Remark 9.6. Here and below C denotes a positive absolute constant whose value may change between occurrences, while each numbered constant C_1, C_2, C_3 denotes a single specific absolute constant, produced at its first

appearance and unchanged thereafter. A nontrivial free homotopy class β simultaneously represents a deck vector $\beta \in \mathbb{Z}^2 - \{0\}$, and $L_\beta = |\beta|$, the Euclidean norm.

The unit-speed geodesic flow of g' is the Hamiltonian flow of $H'(x, p) = \frac{1}{2}F'^*(x, p)^2$ on $\Sigma' = \{H' = \frac{1}{2}\} \subset T^*T^2$, where F'^* is the dual norm; a periodic orbit of period T projects to a closed geodesic of length T . The fiberwise Legendre dual costs at most one derivative, so

$$\|H' - H\|_{C^{k_0-1}(A)} \leq C\delta \quad \text{on } A = \{\tfrac{1}{4} \leq H \leq 1\}, \quad H(p) = \tfrac{1}{2}|p|^2;$$

we rename $k_0 - 1 \rightarrow k_0$ and, enlarging δ by this fixed factor, assume $\|H' - H\|_{C^{k_0}(A)} \leq \delta$.

The flat system is integrable: $\dot{x} = p, \dot{p} = 0$. Its unit level $\Sigma = \{H = \frac{1}{2}\}$ is foliated by the invariant tori $T_\psi = \{p = (\cos \psi, \sin \psi)\}$, $\psi \in S^1$; on T_ψ the flow is linear with frequency vector $(\cos \psi, \sin \psi)$. Here and for the rest of the section $S^1 := \mathbb{R}/2\pi\mathbb{Z}$ denotes the angle circle, of total measure 2π —in contrast with the parametrizing circle $\mathbb{R}/\mathbb{Z}$ of loops used elsewhere in the paper. For δ small, $D_p^2 H'$ is close to the identity, so each locus $\{p : H'(x, p) = \frac{1}{2}\}$ is a strictly convex curve enclosing the origin; it is therefore the radial graph of a function $\rho(x, \cdot) : S^1 \rightarrow \mathbb{R}_{>0}$ with $\|\rho - 1\|_{C^1} \leq C\delta$. In particular,

$$\Psi : \Sigma' \rightarrow T^2 \times S^1, \quad (x, p) \mapsto (x, \psi := \arg p)$$

is a diffeomorphism. For $(x, p) \in \Sigma'$, with $\psi = \arg p$ as above, the radial graph description of the fiber says exactly that $p = \rho(x, \psi)(\cos \psi, \sin \psi)$; hence

$$|p - (\cos \psi, \sin \psi)| = |\rho(x, \psi) - 1| \leq C\delta.$$

9.2. The KAM input. For a real number $r > 0$, call $\omega \in \mathbb{R}^2$ $(r, 2)$ -*Diophantine* if $|\langle \gamma, \omega \rangle| \geq r|\gamma|^{-2}$ for every integer vector $\gamma \in \mathbb{Z}^2 - \{0\}$, and call $\psi \in S^1$ *good* if $(\cos \psi, \sin \psi)$ is $(r, 2)$ -Diophantine. The notion of good depends on the parameter r , which the notation suppresses; r is an adjustable parameter, chosen small in the proof of Proposition 9.1.

Theorem 9.3 (Isoenergetic KAM). *There is an absolute k_0 such that for every $r, \eta > 0$ there is $\delta_0 = \delta_0(r, \eta) > 0$ with the following property. If $\|H' - H\|_{C^{k_0}(A)} \leq \delta_0$, then for every good ψ_0 there is a 2-torus $T'_{\psi_0} \subset \Sigma'$, invariant under the H' -flow, of the form $T'_{\psi_0} = \{(x, P(x)) : x \in T^2\}$ with $\sup_x |P(x) - (\cos \psi_0, \sin \psi_0)| \leq \eta$.*

This is the isoenergetic KAM theorem, in the graph form in which the surviving tori are labelled by their frequency rays; see Arnold [6] for the analytic case, Broer–Huiteima [15] for the derivation of the isoenergetic statement from the ordinary one, and Pöschel [41, 42] for quantitative versions in finite smoothness. The one hypothesis to verify is isoenergetic nondegeneracy of H , i.e. the nonvanishing of the bordered determinant, which here is immediate:

$$\det \begin{pmatrix} D^2 H & \nabla H \\ \nabla H^\top & 0 \end{pmatrix} = -|p|^2 = -1 \quad \text{on } \{H = \tfrac{1}{2}\}.$$

Alternatively—and this is the source of the absolute constant k_0 —on each of four sectors of S^1 on which a coordinate velocity is bounded away from zero, the flow on Σ' reduces to an exact area-preserving twist map of an annulus whose twist is, up to a positive factor, the bordered determinant; Moser's invariant curve theorem [38], in Herman's optimal-smoothness form [26], then produces the invariant tori, and C^1 -closeness to the flat sections makes them automatically graphs.

Lemma 9.4 (Good directions are dense at scale r). *There is an absolute constant C_1 such that, for every $r > 0$, $\text{Leb}(S^1 - \{\text{good}\}) \leq C_1 r$. Since the good set is closed, every complementary gap has length $\leq C_1 r$. In particular, when $C_1 r < 2\pi$ the good set is nonempty and every interval in S^1 of length $> C_1 r$ contains a good angle.*

Proof. The good set is the complement of the union of the sets

$$B_\gamma = \{\psi \in S^1 : |\langle \gamma, (\cos \psi, \sin \psi) \rangle| < r|\gamma|^{-2}\},$$

over integer vectors $\gamma \in \mathbb{Z}^2 - \{0\}$. Each B_γ is open, so the good set is closed. Writing $\gamma = |\gamma|(\cos \theta_\gamma, \sin \theta_\gamma)$, we have $\langle \gamma, (\cos \psi, \sin \psi) \rangle = |\gamma| \cos(\psi - \theta_\gamma)$, so that

$$B_\gamma = \{\psi \in S^1 : |\cos(\psi - \theta_\gamma)| < r|\gamma|^{-3}\}.$$

For $0 \leq s \leq 1$ the set $\{\phi \in S^1 : |\cos \phi| < s\}$ is the union of two arcs of length $2 \arcsin(s)$, centered at $\pm \pi/2$, and $\arcsin(s) \leq \frac{\pi}{2} s$ by convexity of $\arcsin$ on $[0, 1]$; hence

$$\text{Leb}(B_\gamma) \leq 2\pi r |\gamma|^{-3},$$

trivially also when $r|\gamma|^{-3} > 1$. The number of integer vectors γ with $n \leq |\gamma| < n+1$ is at most Cn , so

$$\text{Leb}(S^1 - \{\text{good}\}) \leq \sum_{\gamma \neq 0} \text{Leb}(B_\gamma) \leq 2\pi r \sum_{n \geq 1} \frac{Cn}{n^3} \leq C_1 r.$$

Finally, a complementary gap is a connected component of the open set $S^1 - \{\text{good}\}$, so its length is at most the measure $C_1 r$ of that set. When $C_1 r < 2\pi$ the complement cannot be all of S^1 , so the good set is nonempty, and an interval of length $> C_1 r$ cannot lie in a single gap, so it contains a good angle. $\square$

Lemma 9.5 (Confinement). *For r and η sufficiently small and $\delta \leq \delta_0(r, \eta)$, along every orbit of the H' -flow on Σ' the momentum angle $\psi(t)$ stays in a fixed arc of length $W := C_2(r + \eta)$, where one may take $C_2 = 4C_1 + 24$.*

Proof. In the coordinates Ψ , each T'_{ψ_0} is the graph of $\Theta_{\psi_0}(x) = \arg P(x)$, and for $\eta \leq 1/4$ we have $|\Theta_{\psi_0}(x) - \psi_0| \leq 4\eta$: indeed $|P(x)| \geq 1 - \eta$, so the sine of the angle between $P(x)$ and $(\cos \psi_0, \sin \psi_0)$ is at most $\eta/(1 - \eta)$, and hence the angle is at most 4η , using $\theta \leq \frac{\pi}{2} \sin \theta$ for $\theta \in [0, \pi/2]$. If the orbit starts on a torus it stays there and ψ oscillates by at most 8η . Otherwise, the two intervals below are arcs of length $2C_1 r$, disjoint since r and η are small, so by Lemma 9.4 we may choose good angles

$$\psi_1 \in (\psi(0) - 2C_1 r - 8\eta, \psi(0) - 8\eta), \quad \psi_2 \in (\psi(0) + 8\eta, \psi(0) + 2C_1 r + 8\eta);$$

then pointwise $\Theta_{\psi_1} < \psi(0) < \Theta_{\psi_2}$ and $\Theta_{\psi_1} < \Theta_{\psi_2}$. The band $R = \{(x, \psi) : \Theta_{\psi_1}(x) < \psi < \Theta_{\psi_2}(x)\}$ has $\partial R = T'_{\psi_1} \sqcup T'_{\psi_2}$. If the orbit ever left $\bar{R}$, at the first boundary time it would lie on an invariant torus, forcing the entire orbit—including its initial point—onto that torus, a contradiction. So $\psi(t)$ lies in the arc $[\psi_1 - 4\eta, \psi_2 + 4\eta]$, of length at most $4C_1 r + 24\eta \leq (4C_1 + 24)(r + \eta)$. $\square$

This is where two degrees of freedom enter: the tori have codimension one in $\Sigma' \cong T^2 \times S^1$, and a pair of graph-tori separates it. The graph property here is free (KAM tori are C^1 -small graphs); the intrinsic statements are the Birkhoff-type graph theorems for Lagrangian invariant tori [14, 40, 5] and, for merely continuous incompressible tori of geodesic flows on T^2 , Bialy's theorem [12].

9.3. Proof of Proposition 9.1.

Proof. Length bound. Let o be a closed g' -geodesic of class β . As a Hamiltonian trajectory on Σ' in the Ψ coordinates this gives a trajectory $(x(t), p(t))$ of period $T = \ell_{g'}(o)$, whose lift satisfies $\tilde{x}(T) - \tilde{x}(0) = \beta$, so $\frac{1}{T} \int_0^T \dot{x} dt = \beta/T$ (recalling that β also denotes the corresponding deck vector); in particular $\beta \neq 0$, as the class is nontrivial. Now

$$\dot{x}(t) = \partial_p H'(x, p) = p(t) + O(\delta) = (\cos \psi(t), \sin \psi(t)) + O(\delta),$$

using $|p - (\cos \psi, \sin \psi)| \leq C\delta$. By Lemma 9.5, $\psi(t)$ lies in an arc I of length W . Set

$$e_* := \beta/|\beta|, \quad \varphi_* := \arg e_*,$$

the direction of the total displacement. We claim φ_* lies within distance $4(W + C\delta)$ of I . Indeed, fix any $\psi^0 \in I$; then $|\dot{x}(t) - (\cos \psi^0, \sin \psi^0)| \leq W + C\delta$ for every t , hence the same bound holds for the average β/T , and, as in the proof of Lemma 9.5, a vector within distance $s \leq 1/4$ of a unit vector makes angle at most $4s$ with it (smallness of $W + C\delta$ is arranged by the choice of r, η, δ below). Consequently $|\psi(t) - \varphi_*| \leq 5W + C\delta$ for every t , and since $\psi \mapsto (\cos \psi, \sin \psi)$ is 1-Lipschitz,

$$|\dot{x}(t) - e_*| \leq C_3(W + \delta) =: W' \quad \text{for all } t.$$

Since $|e_*| = 1$, this gives $\langle \dot{x}(t), e_* \rangle \geq 1 - W'$ pointwise, so that

$$T(1 - W') \leq \int_0^T \langle \dot{x}(t), e_* \rangle dt = \left\langle \int_0^T \dot{x}(t) dt, e_* \right\rangle = \langle \beta, \beta/|\beta| \rangle = |\beta| = L_\beta :$$

pairing with the direction of the displacement evaluates its length exactly, with no inequality on the right. Whence $T \leq L_\beta(1 - W')^{-1}$. Choosing r, η with $C_3C_2(r + \eta) \leq \varepsilon/4$ and then $\delta \leq \min\{\delta_0(r, \eta), \varepsilon/(4C_3)\}$, we get $W' \leq \varepsilon/2$, and since $(1 - \varepsilon/2)^{-1} \leq 1 + \varepsilon$ for $\varepsilon \in (0, 1]$, the bound (9.2) follows. The confinement constrains all unit-speed geodesics simultaneously, so this holds for every class β with the same δ .

Tautness and the deformation. By (9.2), $S(g', \beta) = S_{(1+\varepsilon)L_\beta}(g', \beta)$, which is compact by Lemma 3.2; so g' is taut. For g', g'' in the δ -ball, interpolate the squared norms, $F_t^2 = (1 - t)F'^2 + tF''^2$: strong convexity and 2-homogeneity are preserved, and by convexity of the ball each g_t satisfies (9.2). Hence for each β the length spread satisfies

$$\sup_{t \in [0, 1]} \left(\sup_{S(g_t, \beta)} \ell_{g_t} - \inf_{S(g_t, \beta)} \ell_{g_t} \right) \leq (1 + \varepsilon) L_\beta < \infty,$$

and Theorem 2.5 shows $\{g_t\}$ is β -taut for every β , i.e. a taut deformation. Taking $g'' = g_0$ gives the last clause. $\square$

Remark 9.6. Dimension two is essential and enters only in Lemma 9.5: on T^n , $n \geq 3$, KAM tori have codimension $n - 1 \geq 2$ in the energy level and separate nothing; the analogue of Proposition 9.1 is open there and adjacent to Arnold diffusion. Nothing soft can replace the confinement: pointwise bounds $|K| \leq \delta$ do not exclude long class β geodesics—a τ_β -periodic plane curve of geodesic curvature $\leq \delta$ can execute U-turns at length cost $O(1/\delta)$ apiece—so the dynamical confinement does real work. Finally, the proof used only that H is fiberwise quadratic-convex and x -independent with nonvanishing bordered determinant, so it applies verbatim to any constant-coefficient g_0 (as used in the normalization above) and, more generally, to g_0 any flat Minkowski (locally Minkowski Finsler) metric, whose indicatrix replaces the round circle and supplies the radial-graph property, the transversality in Lemma 9.4, and the twist.

10. ARITHMETIC OF CLOSED GEODESICS ON TORI

Proof of Theorem 1.4. By Proposition 9.1 (say with $\varepsilon = 1$), every g in a C^{k_0} δ -ball of the flat metric is β -taut for every nontrivial β , and is joined to the flat metric by a β -taut deformation. For the flat metric itself: after pulling back by a diffeomorphism of T^2 induced by a change of basis of $\mathbb{Z}^2$, we may take β to be a multiple of the class of the first factor of $T^2 = S^1 \times S^1$, and g_0 is joined to a product flat metric by a linear path of flat metrics, which is a β -taut deformation as each flat metric is β -taut; for the product flat metric Theorem 7.1 gives $F = 0$, as $\chi(S^1) = 0$. Since F is a taut deformation invariant, Theorem 2.7, $F(g, \beta) = 0$. $\square$

Lemma 10.1. *Let o be a primitive closed geodesic of a Riemann-Finsler metric on T^2 whose iterates o^k , $k \leq K$, are all nondegenerate, with linearized return map $\mathcal{R} = \mathcal{R}_o \in \text{Sp}(2, \mathbb{R})$. Then $\mathcal{R}$ is positive hyperbolic (eigenvalues $\lambda^{\pm 1}$, $\lambda > 1$), inverse-hyperbolic (eigenvalues $-\lambda^{\pm 1}$, $\lambda > 1$) or elliptic*

(eigenvalues $e^{\pm i\theta}$), and for $k \leq K$:

$$\text{sign det}(I - \mathcal{R}^k) = \begin{cases} -1 & \mathcal{R} \text{ positive hyperbolic,} \\ +1 & \mathcal{R} \text{ elliptic,} \\ (-1)^{k+1} & \mathcal{R} \text{ inverse-hyperbolic.} \end{cases}$$

Proof. Nondegeneracy of o and o^2 excludes the eigenvalue pairs $\{1, 1\}$ and $\{-1, -1\}$, giving the trichotomy. If the eigenvalues are $\lambda^{\pm 1}$, $\lambda > 1$, then

$$\det(I - \mathcal{R}^k) = (1 - \lambda^k)(1 - \lambda^{-k}) < 0.$$

If they are $e^{\pm i\theta}$, then

$$\det(I - \mathcal{R}^k) = |1 - e^{ik\theta}|^2 > 0,$$

nonzero by nondegeneracy of o^k . If they are $-\lambda^{\pm 1}$, $\lambda > 1$, then

$$\det(I - \mathcal{R}^k) = (1 - (-\lambda)^k)(1 - (-\lambda)^{-k}),$$

positive for k odd and negative for k even. $\square$

Proof of Proposition 1.5. Tautness and regularity make each set of class $d\beta_0$ strings finite. Every closed geodesic string is the k -fold cover o^k of a unique primitive geodesic string o , and o^k has multiplicity k ; if o has class $e\beta_0$ then o^k has class $ke\beta_0$. So the class $d\beta_0$ strings are in bijection with the pairs (e, o) , where $e|d$ and $o \in P_e$, the pair (e, o) corresponding to $o^{d/e}$, of multiplicity d/e . By the definition of F and Appendix A, the contribution of the string $o^{d/e}$ to $F(g, d\beta_0)$ is its local Fuller index divided by its multiplicity, that is $i(o^{d/e})/(d/e)$ with i as above:

$$F(g, d\beta_0) = \sum_{e|d} \sum_{o \in P_e} \frac{i(o^{d/e})}{d/e},$$

and multiplying by d , so as to obtain an identity between integers,

$$(10.2) \quad dF(g, d\beta_0) = \sum_{e|d} e \sum_{o \in P_e} i(o^{d/e}),$$

since $d/(d/e) = e$.

Now let u_e be the number of positive hyperbolic minus the number of elliptic elements of P_e , so that $s_e = u_e - b_e$ and, by Lemma 10.1, $\sum_{o \in P_e} i(o^k) = u_e + (-1)^k b_e$ for every k . Since $(-1)^{d/e} = 1$ exactly when d/e is even, that is when e divides $d/2$, substitution in (10.2) gives

$$dF(g, d\beta_0) = \sum_{e|d} e(u_e - b_e) + 2 \sum_{e|d/2} e b_e = \sum_{e|d} e s_e + 2 \sum_{e|d/2} e b_e,$$

the second sum being empty for d odd. With $B_x := \sum_{e|x} e b_e$, and $B_x := 0$ when x is not an integer, the vanishing of $F(g, d\beta_0)$ for all d reads

$$(10.3) \quad \sum_{e|d} e s_e = -2B_{d/2} \quad \text{for all } d \geq 1.$$

Now for any d ,

$$(10.4) \quad \sum_{e|d} e b_{e/2} = \sum_{2f|d} 2f b_f = 2B_{d/2},$$

since the terms with e odd vanish, and the even divisors of d are exactly $e = 2f$ with f a divisor of $d/2$. This gives the converse direction at once: if $s_e = -b_{e/2}$ for every e , then the left hand side of (10.3) is the negative of that of (10.4), so (10.3) holds. For the forward direction argue by induction

on d , the case $d = 1$ of (10.3) being $s_1 = -2B_{1/2} = 0$. Assuming $s_e = -b_{e/2}$ for every proper divisor e of d , (10.3) and (10.4) give

$$d s_d = -2B_{d/2} + \sum_{e|d, e \neq d} e b_{e/2} = -2B_{d/2} + (2B_{d/2} - d b_{d/2}) = -d b_{d/2},$$

hence $s_d = -b_{d/2}$. $\square$

APPENDIX A. FULLER INDEX AND SKY CATASTROPHES

In this appendix we recall the part of Fuller's periodic orbit index used in the paper, and fix the notation for orbit strings and for continuation under small perturbations of vector fields. Some of this material overlaps with [46]; we nevertheless give a brief self-contained account, especially since the noncompact setting needed here is not treated there, and the details serve to fix notation used throughout the paper.

Let X be a smooth manifold, and fix a complete auxiliary Riemannian metric h on X . Let

$$\mathcal{V}(X)$$

denote the space of smooth vector fields on X , with the C^0 -topology on compact subsets.

Let

$$LX = C^\infty(S^1, X)$$

denote the smooth free loop space. For $V \in \mathcal{V}(X)$, define

$$S(V) = \{o \in LX \mid o \text{ is nonconstant, } \dot{o}(t) = T V(o(t)) \text{ for some } T > 0\}.$$

We call it the period of o , and write

$$T = T(o),$$

it is clearly uniquely determined.

The circle $S^1 = \mathbb{R}/\mathbb{Z}$ acts on $S(V)$ by reparametrization:

$$(s \cdot o)(t) = o(t + s).$$

We define the space of orbit strings by

$$\mathcal{O}(V) = S(V)/S^1.$$

If β is a free homotopy class in X , we write

$$S(V, \beta) \subset S(V), \quad \mathcal{O}(V, \beta) \subset \mathcal{O}(V)$$

for the corresponding subspaces. When no confusion is possible, we use the same letter o for a parametrized orbit and for its associated orbit string.

Let

$$q : LX \rightarrow LX/S^1$$

be the quotient map. If

$$U \subset LX/S^1,$$

we write

$$\hat{U} = q^{-1}(U) \subset LX.$$

Definition A.1. We say that U is *h -controlled* if there are constants

$$0 < \ell_- < \ell_+ < \infty$$

and a compact set $K \subset X$ such that every loop $o \in \hat{U}$ satisfies

$$\text{image}(o) \subset K, \quad \ell_- \leq \ell_h(o) \leq \ell_+.$$

The lower bound excludes collapse to constant loops; in the applications it is usually supplied by a nontrivial projected free homotopy class.

For an orbit string $o \in \mathcal{O}(V)$, its multiplicity

$$m(o) \in \mathbb{N}$$

is the unique positive integer such that o is the $m(o)$ -fold cover of a primitive orbit string. Equivalently, if $T(o)$ is the period of o and $T_{\text{prim}}(o)$ is the period of the underlying primitive orbit string, then

$$m(o) = \frac{T(o)}{T_{\text{prim}}(o)}.$$

A.1. Fuller index. Suppose

$$N \subset \mathcal{O}(V)$$

is finite and consists of nondegenerate orbit strings. Fuller associates to (N, V) the rational number

$$i(N, V) = \sum_{o \in N} \frac{i(o)}{m(o)}.$$

Here, for a nondegenerate orbit string o of the flow of V , on an ambient manifold of dimension $2m+1$, and with linearized return map $\mathcal{R}_o$ computed on a local hypersurface transverse to the image of o , we set

$$(A.2) \quad i(o) := (-1)^m \text{sign det}(I - \mathcal{R}_o).$$

Up to the dimensional sign $(-1)^m$ this is the fixed point index of the time- $T(o)$ return map; the sign is chosen so that minimizing closed geodesics contribute positively, see below.

By a compact open subset N of an orbit-string space, we mean a subset which is compact and open in the relative topology. In this case Fuller again associates to this a rational index $i(N, V)$. The latter can be defined perturbatively, and using the formula as above.

In the special case of a nondegenerate closed geodesic orbit, with ambient manifold $S_g X$ so that $m = \dim X - 1$, the Morse index theorem gives the classical relation $\text{sign det}(I - \mathcal{R}_o) = (-1)^{\text{morse}(o)+m}$, see for example [31, Chapter 12], so that by (A.2):

$$i(\tilde{o}) = (-1)^{\text{morse}(o)}.$$

Consequently, if N consists of finitely many nondegenerate closed orbits of the geodesic Reeb flow R^{λ_g} on C , we get:

$$i(N, V) = \sum_{o \in N} \frac{(-1)^{\text{morse}(o)}}{m(o)}.$$

A.2. Continuation. Let

$$\{V_t\}_{t \in [0,1]} \subset \mathcal{V}(X)$$

be a continuous family. Define

$$S(\{V_t\}) = \{(o, t) \in LX \times [0, 1] \mid o \in S(V_t)\},$$

and

$$(A.3) \quad \mathcal{O}(\{V_t\}) = S(\{V_t\})/S^1,$$

where S^1 acts on the loop coordinate by reparametrization as before.

Let

$$\mathcal{N} \subset \mathcal{O}(\{V_t\})$$

be compact and open. Suppose that its endpoint slices are

$$N_0 = \mathcal{N} \cap \mathcal{O}(V_0), \quad N_1 = \mathcal{N} \cap \mathcal{O}(V_1).$$

Then

$$(A.4) \quad i(N_0, V_0) = i(N_1, V_1).$$

We refer to this as the **basic invariance** of the Fuller index.

For applications, we need a local version of this statement under small perturbations. This is a variation of [23, Lemma 4.1], with the added control needed for noncompact X .

Lemma A.5 (Isolation lemma). *Let h be a complete auxiliary Riemannian metric on X . Let $V_0 \in \mathcal{V}(X)$, be a nonsingular vector field, and let*

$$N_0 \subset \mathcal{O}(V_0)$$

be compact and open. Let

$$U \subset LX/S^1$$

be an h -controlled open neighborhood of N_0 such that

$$\overline{U} \cap \mathcal{O}(V_0) = N_0.$$

Let W be an open neighborhood of N_0 satisfying

$$\overline{W} \subset U.$$

Then there exists $\epsilon > 0$ with the following property. If V is a smooth vector field satisfying

$$\sup_{x \in K} |V(x) - V_0(x)|_h < \epsilon,$$

where K is the compact set from the h -control of U , then

$$\mathcal{O}(V) \cap U \subset W.$$

In particular,

$$\mathcal{O}(V) \cap U$$

is compact and open in $\mathcal{O}(V)$.

Proof. Let K , ℓ_- , and ℓ_+ be the compact set and length bounds from the h -control of U . Since V_0 is nonsingular and K is compact, there are constants

$$0 < B < A < \infty$$

such that

$$B \leq |V_0(x)|_h \leq A$$

for all $x \in K$. We choose $\epsilon < B/2$. Then every vector field V satisfying

$$\sup_K |V - V_0|_h < \epsilon$$

obeys

$$|V(x)|_h \geq B/2, \quad |V(x)|_h \leq A + \epsilon$$

for all $x \in K$.

Suppose the conclusion failed for the chosen W . Then there would be vector fields

$$V_j \rightarrow V_0$$

uniformly on K , and orbit strings

$$o_j \in (\mathcal{O}(V_j) \cap U) - W.$$

Choose parametrized representatives, still denoted o_j , satisfying

$$(A.6) \quad \dot{o}_j(s) = T_j V_j(o_j(s)).$$

Since $o_j \in \widehat{U}$, controlledness gives

$$\text{image}(o_j) \subset K, \quad \ell_- \leq \ell_h(o_j) \leq \ell_+.$$

For j sufficiently large,

$$B/2 \leq |V_j|_h \leq A + \epsilon$$

on K . Hence

$$\frac{\ell_-}{A + \epsilon} \leq T_j \leq \frac{2\ell_+}{B}.$$

The velocities are also uniformly bounded:

$$|\dot{o}_j(s)|_h = T_j |V_j(o_j(s))|_h \leq \frac{2\ell_+(A + \epsilon)}{B}.$$

Thus the o_j are equicontinuous and have images in the fixed compact set K . By the Arzelà–Ascoli theorem, after passing to a subsequence,

$$o_j \rightarrow o_\infty$$

uniformly. After passing to a further subsequence, $T_j \rightarrow T_\infty$, with $T_\infty > 0$.

Hence o_j is a Cauchy sequence in the C^1 space of maps, by the defining equation (A.6). Then C^1 completeness gives that

$$\dot{o}_\infty = T_\infty V_0(o_\infty).$$

Thus o_∞ is a closed V_0 -orbit string. Moreover

$$o_\infty \in \overline{U} - W.$$

This contradicts

$$\overline{U} \cap \mathcal{O}(V_0) = N_0 \subset W.$$

Therefore

$$\mathcal{O}(V) \cap U \subset W$$

for all V sufficiently close to V_0 on K .

It remains to prove compactness. Let o_j be any sequence in $\mathcal{O}(V) \cap U$. The same estimates as above give uniformly bounded periods and velocities, with all images contained in K . Arzelà–Ascoli therefore gives a convergent subsequence. Its limit is again a closed V -orbit. Since the inclusion just proved gives

$$\mathcal{O}(V) \cap U = \mathcal{O}(V) \cap \overline{W}$$

and $\overline{W} \subset U$, the limit lies in $\mathcal{O}(V) \cap U$. Thus $\mathcal{O}(V) \cap U$ is compact. It is open because U is open. $\square$

We call such an ϵ a **Fuller constant** for the data

$$(V_0, N_0, U, W).$$

It also depends on the auxiliary control data K, ℓ_-, ℓ_+ , which will always be understood from the chosen U .

Lemma A.7 (Fuller continuation). *Let V_0, N_0, U, W , and ϵ be as in Lemma A.5. Let*

$$\{V_t\}_{t \in [0,1]}, \quad V_{t=0} = V_0$$

be a continuous family such that for every $t \in [0,1]$,

$$\sup_{x \in K} |V_t(x) - V_0(x)|_h < \epsilon,$$

where K is the compact set from the h -control of U . Define

$$\mathcal{N} = (U \times [0,1]) \cap \mathcal{O}(\{V_t\}).$$

Then $\mathcal{N}$ is compact and open in the orbit cobordism $\mathcal{O}(\{V_t\})$. Its endpoint slices are

$$N_0 = U \cap \mathcal{O}(V_0), \quad N_1 = U \cap \mathcal{O}(V_1).$$

Consequently,

$$i(N_0, V_0) = i(N_1, V_1).$$

We call N_1 the Fuller continuation of N_0 .

Proof. Since U is open,

$$\mathcal{N} = (U \times [0, 1]) \cap \mathcal{O}(\{V_t\})$$

is open in $\mathcal{O}(\{V_t\})$.

By Lemma A.5, applied to each V_t , we have

$$\mathcal{O}(V_t) \cap U \subset W$$

for every $t \in [0, 1]$. Since

$$\overline{W} \subset U,$$

this implies

$$\mathcal{N} = (\overline{W} \times [0, 1]) \cap \mathcal{O}(\{V_t\}).$$

The compactness then works as in Lemma A.5. The endpoint slices are

$$\mathcal{N} \cap \mathcal{O}(V_0) = U \cap \mathcal{O}(V_0) = N_0$$

and

$$\mathcal{N} \cap \mathcal{O}(V_1) = U \cap \mathcal{O}(V_1) = N_1.$$

Fuller's basic invariance applied to the compact open cobordism $\mathcal{N}$ gives

$$i(N_0, V_0) = i(N_1, V_1).$$

□

Example 7 (Geodesic flows and the Sasaki metric). Let (Y, g_Y) be a complete Riemannian manifold, and let

$$C_g = S^*Y$$

be its unit cotangent bundle with Liouville contact form λ_{g_Y} . The Reeb vector field $R^{\lambda_{g_Y}}$ is the geodesic Reeb field.

Let g_S be the Sasaki metric on S^*Y . It is defined using the Levi-Civita connection of g_Y . Thus

$$TC = T^{\text{vert}}C \oplus T^{\text{hor}}C$$

is an orthogonal splitting, where

$$T^{\text{vert}}C = \ker(\text{pr}_*)$$

and $T^{\text{hor}}C$ is the Levi-Civita horizontal subbundle. If $\xi, \eta \in T(S^*Y)$ are horizontal, then

$$g_S(\xi, \eta) = g_Y(\text{pr}_* \xi, \text{pr}_* \eta),$$

and the vertical part is measured using the metric induced by g_Y on the cotangent fibers.

The basic property needed in the paper is the following. If o is a nonzero constant-speed g_Y -geodesic and $\tilde{o}$ is its unit cotangent lift, then

$$(A.8) \quad \ell_{g_S}(\tilde{o}) = \ell_{g_Y}(o).$$

Indeed, the covector component of $\tilde{o}$ is parallel along o , so the vertical part of $\tilde{o}$ vanishes and the Sasaki speed equals the base speed.

If $\beta \in \pi_1^{\text{inc}}(Y)$ and $U \subset S(g_Y, \beta)$ is g_Y -length bounded, then the set of canonical lifts

$$\tilde{U} \subset S(R^{\lambda_{g_Y}})$$

is g_S -length bounded by (A.8).

APPENDIX B. A LOCAL FIBERED CALCULATION OF F

We prove in this section the technical lemma computing the local index, used in Theorem 7.1.

Let

$$P : S^*X \rightarrow \mathbb{R}$$

be the verticality function as in the proof of Theorem 7.1,

$$P(\xi) = |\pi^{\text{vert}} v_\xi|_g^2.$$

Here v_ξ is the g -dual unit vector to ξ , and

$$\pi^{\text{vert}} : TX \rightarrow T^{\text{vert}}X$$

is the g -orthogonal projection. The notation β and β_Z is likewise as in Theorem 7.1.

Lemma B.1 (Local product calculation). *Let*

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a parallel Riemannian submersion. Let

$$\tilde{f} = f \circ p \circ \text{pr}$$

*for a Morse function $f : Y \rightarrow \mathbb{R}$, and define the vector field on S^*X*

$$V_\varepsilon = R^{\lambda_g} - \varepsilon \nabla_{g_S}(P + \tilde{f}).$$

Fix

$$y \in \text{Crit}(f),$$

and let

$$N_{\varepsilon, y}$$

be the isolated set of closed V_ε -orbit strings lying over the fiber

$$Z_y = p^{-1}(y),$$

as in the proof of Theorem 7.1. Then

$$i(N_{\varepsilon, y}, V_\varepsilon) = (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha),$$

where S_y is the $\pi_1(Y, y)$ -orbit of the corresponding fiber representative in $\pi_1^{\text{inc}}(Z_y)$. In particular, if the fibration is holonomy-finite in the relevant class, the sum is finite.

Proof. Let

$$T^{\text{vert}}X = \ker(dp), \quad H = (T^{\text{vert}}X)^{\perp_g}.$$

Since the submersion is parallel, the two distributions

$$T^{\text{vert}}X \quad \text{and} \quad H$$

are preserved by the Levi-Civita connection. Since $T^{\text{vert}}X$ and $H = (T^{\text{vert}}X)^{\perp_g}$ are parallel complementary distributions, the local de Rham decomposition theorem gives a product decomposition

$$p^{-1}(D_y) \cong Z_y \times D_y, \quad g = g_y \oplus g_Y|_{D_y},$$

for a normal ball neighborhood $D_y \subset Y$ centered at y . See Kobayashi–Nomizu [29, Chapter IV].

For this local calculation we suppress the fixed- C identification and write S^*X for the g -unit cotangent bundle. Let

$$C^{\text{vert}} = \{P = 1\} \subset S^*X$$

be the vertical locus. Over Z_y , this locus is naturally identified with

$$S^*Z_y \subset S^*X.$$

Near

$$S^*Z_y \subset C^{\text{vert}},$$

the product decomposition gives the normal-bundle model

$$(B.2) \quad S^*Z_y \times D_y \times B_y^*,$$

where

$$B_y^* \subset T_y^*Y$$

is a small ball in the horizontal cotangent direction. In this model, the vertical locus is

$$S^*Z_y \times D_y \times \{0\}.$$

A covector near the vertical locus decomposes as

$$\xi = \xi_{\text{vert}} + \eta, \quad \eta \in B_y^*.$$

The unit condition is

$$|\xi_{\text{vert}}|_{g_y^*}^2 + |\eta|_{g_Y^*}^2 = 1.$$

Therefore

$$P(\xi) = |\xi_{\text{vert}}|_{g_y^*}^2 = 1 - |\eta|_{g_Y^*}^2.$$

Thus P has a Morse–Bott maximum along C^{vert} .

In the local model (B.2),

$$N_{\varepsilon,y} = N_y^{\text{fib}} \times \{y\} \times \{0\},$$

where N_y^{fib} is the relevant closed orbit set of the fiber geodesic Reeb flow on S^*Z_y .

By a C^∞ -small perturbation of g_y , under which both sides of the claimed identity are unchanged by the invariance of the Fuller index, we may assume the fiber strings are nondegenerate, and hence finite in number.

We now describe the local return map at an element of $N_{\varepsilon,y}$. Along the vertical locus over Z_y , the fiber component of V_ε is the geodesic Reeb field $R^{\lambda_{g_y}}$ on S^*Z_y . The base component is

$$-\varepsilon \nabla_{g_Y} f$$

on D_y . The normal covector component is the outward radial component coming from

$$-\varepsilon \nabla_{g_S} P$$

on B_y^* .

Let

$$o \in N_y^{\text{fib}},$$

and choose a product transverse section

$$\Sigma = \Sigma^{\text{fib}} \times D_y \times B_y^*,$$

where

$$\Sigma^{\text{fib}} \subset S^*Z_y$$

is transverse to the fiber Reeb flow near o , and let $T(o)$ be the return time. In the coordinates (fiber, base, normal) of the product section, denote by σ the local return map of V_ε ; it is block triangular over the normal variable η . The normal component is a map κ of B_y^* alone, the time- $T(o)$ map of the radial field induced by $-\varepsilon \nabla_{g_S} P$, while the fiber and base components depend on η but restrict at $\eta = 0$ to, respectively, the local return map Φ_{fib} of the fiber geodesic flow generated by $R^{\lambda_{g_y}}$, and the time- $T(o)$ map ψ_y of $-\varepsilon \nabla_{g_Y} f$ near y .

The fixed point of σ corresponding to o is $(w, y, 0)$, where w is the point where o meets the fiber section. Consequently the linearized return map $\mathcal{R}_o = d\sigma$ at this fixed point is block triangular with diagonal blocks $d\Phi_{\text{fib}}$, $d\psi_y$ and $d\kappa$, so that

$$\det(I - \mathcal{R}_o) = \det(I - d\Phi_{\text{fib}}) \cdot \det(I - d\psi_y) \cdot \det(I - d\kappa);$$

compare also the multiplicativity of the fixed point index under products, Dold [21, Chapter VII, Section 5]. Here, for a map ϕ with an isolated fixed point x , we write $i_{\text{fp}}(\phi, x) \in \mathbb{Z}$ for its fixed point index: the local degree at x of the map $\text{id} - \phi$, computed in any chart; at a nondegenerate fixed point,

$$i_{\text{fp}}(\phi, x) = \text{sign } \det(I - d\phi_x).$$

Hence

$$i_{\text{fp}}(\sigma, (w, y, 0)) = i_{\text{fp}}(\Phi_{\text{fib}}, w) \cdot i_{\text{fp}}(\psi_y, y) \cdot i_{\text{fp}}(\kappa, 0).$$

In normal coordinates at y , the map $\text{id} - \psi_y$ equals $\varepsilon T(o) \nabla_{g_Y} f$ to first order, so has the same local degree as $\nabla_{g_Y} f$. Thus, because y is a Morse nondegenerate critical point,

$$i_{\text{fp}}(\psi_y, y) = (-1)^{\text{morse}(y)}.$$

Since the normal component of

$$-\varepsilon \nabla_{g_S} P$$

has outward radial linearization on a vector space of dimension $\dim Y$, the map $\text{id} - \kappa$ has local degree $(-1)^{\dim Y}$. Hence

$$i_{\text{fp}}(\kappa, 0) = (-1)^{\dim Y}.$$

Therefore

$$i_{\text{fp}}(\sigma, (w, y, 0)) = (-1)^{\dim Y + \text{morse}(y)} i_{\text{fp}}(\Phi_{\text{fib}}, w).$$

The multiplicity of o is unchanged by taking the product with the isolated base and normal fixed points. Finally we pass from fixed point indices to the Fuller weights of (A.2): the ambient manifolds are S^*X on the left and S^*Z_y on the right, of dimensions $2(\dim Z + \dim Y) - 1$ and $2\dim Z - 1$, so the two dimensional signs differ by exactly $(-1)^{\dim Y}$, cancelling the factor above:

$$\frac{i(o; V_\varepsilon)}{m(o)} = (-1)^{\text{morse}(y)} \frac{i(o; R^{\lambda_{g_Y}})}{m(o)}.$$

Summing over all fiber orbit strings in the local block gives

$$i(N_{\varepsilon, y}, V_\varepsilon) = (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha).$$

This proves the claim. $\square$

APPENDIX C. SKY CATASTROPHES

We now recall the topological formulation of sky catastrophes as studied in [46]. This is the formulation necessary for Section 2.0.2.

Definition C.1. Let

$$\{V_t\}_{t \in [0,1]} \subset \mathcal{V}(X)$$

be a continuous family. We say that the family has a sky catastrophe in class β if there is an endpoint orbit string

$$y \in \mathcal{O}(V_0, \beta) \cup \mathcal{O}(V_1, \beta) \subset \mathcal{O}(\{V_t\}, \beta)$$

such that no compact open subset of $\mathcal{O}(\{V_t\}, \beta)$ contains y .

Equivalently, the endpoint orbit string y cannot be included in any compact continuation block through the family.

For suitably regular families, this agrees with the preliminary Definition 2.10.

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